# Group and solo flight: comparisons

Group time against solo flight, raw, on the same route, by factor and around the group, on open flights

Matteo Di Sante

November 2, 2026 • analysis: `xc-lab/analyses/groups` • grouping run `d120_eh500_ez300_th45_g120_q0.1`

**Abstract**

We compare paraglider group time with solo flight on open flights, four ways: raw; against solo flights on the same route, season, hour and wing; by how much each of those factors accounts for the raw gap; and against the same flight before and after its group time. All bands come from a two-way bootstrap of pilots and blocks, whose 90% coverage we check by simulation on the designs of the data. Raw, group time progresses straighter than solo flight at the long lags: $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is 0.915 in G and 0.897 in S, a gap of $+0.018$ [$+0.015$, $+0.021$]. On the same route, season, hour and wing the gap of $H_{\mathrm{loc}}(1000\ \mathrm{s})$ falls to $+0.007$ [$-0.000$, $+0.015$] on pieces of at least 1 250 s, while on longer pieces $H_{\mathrm{loc}}(1840\ \mathrm{s})$ gives $+0.018$ [$+0.004$, $+0.032$]; group time also climbs more often and glides slower and steeper than its controls, and the same pilots alone do not differ from theirs. Taking the factors apart, the route carries the largest part of the raw differences of time in climb, glide speed and height lost per km of glide. Around the group, the drift-free effect $g$ of $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is $+0.026$ [$+0.020$, $+0.032$], close to the same-route gap on the longer pieces. Time in search, climb rate, glide speed and height lost per km of glide have opposite signs in the matched and in the before-and-after comparison.

## 1 Aim

Does a paraglider move differently in a group than alone? This report compares group time with solo flight on open flights in four ways, each on the same populations and measures.

1. **Raw** (Section 5): S, A and G as they are.
2. **Same route** (Section 6): pieces of group time against solo segments on the same route, in the same season and hour, with the same wing.
3. **Decomposition** (Section 7): how much of the raw gap the route, the season and hour, and the wing each account for.
4. **Before, during, after** (Section 8): the same flight in the windows around its group time.

Section 2 defines the flights, the populations and the measures; Sections 3 and 4 say which flights are open and why. Every table and figure gives the flights or pieces it rests on, its dataset and its estimator. Appendices give the formulas of each tool.

## 2 Flights, populations and measures

Which flights, which stretches of flight, and how are they measured?

**Declared open flights.** On the FFVL catalogue the pilot declares the type of the flight (Table 1). A distance, free or with up to three turn points, is open; a triangle, a quadrilateral or an out-and-return is closed; a hike and fly is neither. We keep the declared open flights: 66 845 of the 154 598 paraglider flights that depart and are logged every 10 s or finer (87 484 are closed, 269 neither).

**Groups.** The labels are those of the note *Group flights in free flight* (`methods/group-identification`). Two pilots are in contact when one is on the path of the other within $\delta = 2$ min, $\varepsilon_h = 500$ m horizontally and $\varepsilon_z = 300$ m vertically (headings compared only in glide) for at least $\tau^* = 38.1$ min. The configurations of an instant are the connected components of the contact graph, and a group keeps the same members over a maximal interval. The groups are those of every flight of the archive, whoever the companions.

| label | meaning | class | flights |
|---|---|---|---|
| Dist 3 pts | distance, 3 turn points | open | 40 094 |
| Dist 2 pts | distance, 2 turn points | open | 25 399 |
| Dist libre | free distance | open | 694 |
| Dist 1 pt | distance, 1 turn point | open | 658 |
| triangle | triangle | closed | 51 181 |
| triangle FAI | FAI triangle | closed | 34 979 |
| Quadrilatère | quadrilateral | closed | 703 |
| Aller-Retour | out and return | closed | 617 |
| Marche et Vol bouclé | hike and fly, closed | closed | 3 |
| Marche et Vol FAI | hike and fly, FAI triangle | closed | 1 |
| Marche et Vol | hike and fly | neither | 269 |

**Table 1:** The labels of the FFVL catalogue, their class and their flights (paragliders that depart, logged every 10 s or finer).

**Populations.** After its departure, the first fix 3 km from the takeoff, every instant of a flight is in a group or solo. Three populations:

- S, *solo flights*: flights that never entered a group, the whole of their time after the departure;
- G, *group time*: the stretches in a group of the flights that entered one at least once;
- A, *the same flights alone*: the solo stretches of those same flights, with the same pilots, gliders, loggers and days as G.

**Measure.** A *piece* is a stretch within one segment of the track, on a 10-s grid of its clock. For a flight $f$ and a population, the time-averaged mean squared displacement pools the origins of all its pieces in that population,

$$\overline{\delta_f^2}(\tau) = \frac{1}{N_f(\tau)} \sum_t \left\| \mathbf{r}(t+\tau) - \mathbf{r}(t) \right\|^2, \tag{1}$$

with $t$ and $t+\tau$ in the same piece. The ensemble $\mathrm{MSD}(\tau)$ is its mean over the flights, one weight per flight. $H_{\mathrm{loc}}(\tau)$ is half the slope of $\log \mathrm{MSD}$ against $\log \tau$ over the lags within $\pm 0.25$ decades of $\tau$: 1 for a straight line, $\frac{1}{2}$ for a random walk. $H$ is half the slope over all the lags of a curve, from 10 s to its end. Bands are 90% bands of a two-way bootstrap of pilots and blocks (1 000 draws of each, Appendix H).

**Datasets.** A lag $\tau$ needs pieces of at least $\tau/0.8$ s, so that the origins of every piece cover the lag in the same share of it. A dataset to the lag $T$ keeps, in each population, its pieces of at least $T/0.8$ s and measures them at every lag up to $T$: the population does not change along a curve (Appendix C). Three datasets:

- *1000*: $T = 1\,000$ s, pieces of at least 1 250 s;
- *2250*: $T = 2\,250$ s, pieces of at least 2 812.5 s;
- *longest*: $T = 11\,450$ s, the longest lag at which each population keeps 100 flights, pieces of at least 14 312.5 s.

The quoted values are $H_{\mathrm{loc}}(1000\ \mathrm{s})$ on dataset 1000 and $H_{\mathrm{loc}}(1840\ \mathrm{s})$ on dataset 2250, the lag of the grid nearest to 2000 s. The window of $H_{\mathrm{loc}}(1840\ \mathrm{s})$, $\pm 0.25$ decades, is cut at 2250 s: it uses the lags from 1 035 to 2250 s only.

**Phase statistics.** Flights logged at 1 Hz carry the phases of Vilpellet's method: climbs, searches and glides. Phases exist on 1-Hz flights only, so the phase statistics rest on fewer flights than the MSD (27 665 in S, 6 397 in A and 7 896 in G against 43 977, 8 991 and 11 077 on dataset 1000). A run belongs to the unit that holds its middle. Seven statistics, each a ratio $n/d$ of two terms summed over the runs of a unit (Table 2); the phase time $T_u$ of a unit $u$ is the sum of the durations of its runs.

| statistic | numerator $n$ | denominator $d$ | runs counted | $r_u$, $q_u$ |
|---|---|---|---|---|
| time in climb (%) | time in climbs | phase time | climbs | $n_u/T_u$, constant |
| time in search (%) | time in searches | phase time | searches, found and vain | $n_u/T_u$, constant |
| searches ending in a climb (%) | searches followed by a climb | searches | searches | $n_u/T_u$, $d_u/T_u$ |
| climbs per hour | climbs | phase time (h) | climbs | $n_u/T_u$, constant |
| climb rate (m/s) | height gained in climbs | time in climbs | climbs | $n_u/T_u$, $d_u/T_u$ |
| glide speed (km/h) | straight-line distance of glides | time in glides | glides | $n_u/T_u$, $d_u/T_u$ |
| height lost per km of glide (m/km) | height lost in glides | distance of glides | glides | $n_u/T_u$, $d_u/T_u$ |

**Table 2:** The seven phase statistics. Unit $u$ of the average: the flight in Section 5 (its runs on its pieces of dataset 1000 in the population), the piece or control segment in Sections 6 and 7, the window in Section 8.

**Estimator: average first, ratio after.** With $w_u$ the weight of a unit (1 for the value, the bootstrap count for a draw),

$$r_u = \frac{n_u}{T_u}, \qquad q_u = \frac{d_u}{T_u}, \qquad \text{stat} = \frac{\sum_u w_u\, r_u}{\sum_u w_u\, q_u}. \tag{2}$$

For the two time shares and the climbs per hour $q_u$ is constant, and the statistic is the mean of the unit's share or rate. For the other four it does not divide by the few events of one unit: a unit with one search gives 0% or 100%. This is the estimator of the MSD, one weight per unit. The ratio of sums, $\sum n_u / \sum d_u$, weights a unit by its phase time (Appendix D).

**Survival of the phase durations.** For climbs, searches that end in a climb and glides, the survival function $P(\tau_{\text{phase}} > \tau)$ of the number of fixes of a run (1 Hz, so seconds), with the duration trimmed as in the note on phases. Every run weighs 1; its band holds at each $t$ (Appendix H) (Appendix E).

# 3 Group, solo and alone on the declared open flights

How do the three populations compare as they are, with no stratification?

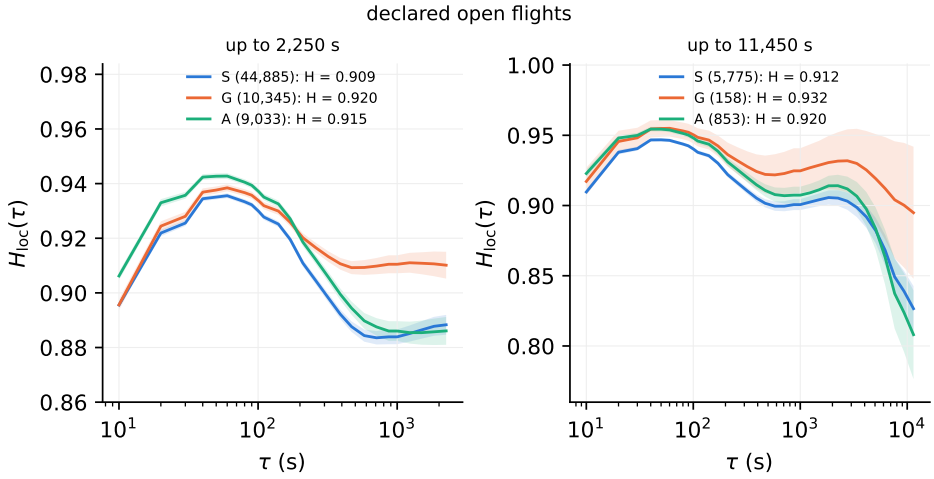



**Figure 1:** Declared open flights. $H_{\text{loc}}(\tau)$ of S, G and A for the datasets up to 2 250 s (left) and 11 450 s (right), each on the pieces of at least $T/0.8$ s, with their flights and the fitted $H$ in the legend; 90% bands. Flights of each curve in Table 3.

| dataset | S | G | A |
|---|---|---|---|
| lags up to 2 250 s | 44 885 | 10 345 | 9 033 |
| lags up to 11 450 s | 5 775 | 158 | 853 |

**Table 3:** Flights of each dataset of Figure 1, declared open flights.

On the dataset up to 2 250 s group time is the straightest (Figure 1): $H_{\text{loc}}(1000\text{ s})$ is 0.910 in G, against 0.884 in S and 0.886 in A, and the fitted $H$ is 0.920, 0.909 and 0.915. On the dataset up to 11 450 s the mean squared displacement of S and A bends down at lags of hours: at 11 450 s $H_{\text{loc}}$ is 0.827 in S and 0.808 in A. That of G stays at 0.895 [0.848, 0.942], on 158 flights only; the fitted $H$ is 0.932, 0.912 and 0.920.

**What we keep.** Raw, group time progresses straighter than solo flights and than the same flights alone, and at lags of hours the motion of solo flights and of the same flights alone bends away from a power law.

# 4 How open the declared open flights are

Why does the mean squared displacement bend down at long lags?

**Opening.** The *opening* of a flight is the distance from its takeoff to its landing, its first and last fix, over the path flown: 1 for a straight line, near 0 for a flight that comes back to its takeoff.

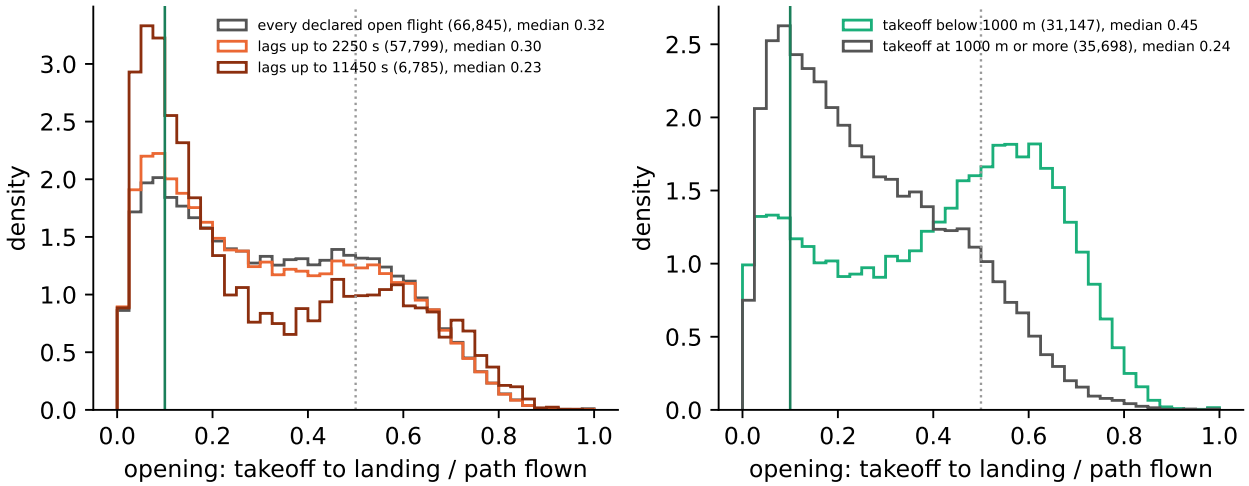



**Figure 2:** Opening of the declared open flights, takeoff to landing over the path flown. Left: every declared open flight and the flights of the datasets of Figure 1. Right: takeoffs below and above 1 000 m. Solid, the threshold 0.1; dotted, the stricter 0.5 that we do not use.

| | flights | | | | median opening | | | |
|---|---|---|---|---|---|---|---|---|
| dataset | all | S | G | A | all | S | G | A |
| every declared open flight | 66 845 | | | | 0.32 | | | |
| lags up to 2 250 s | 57 799 | 44 885 | 10 345 | 9 033 | 0.30 | 0.31 | 0.28 | 0.23 |
| lags up to 11 450 s | 6 785 | 5 775 | 158 | 853 | 0.23 | 0.23 | 0.29 | 0.19 |

**Table 4:** Flights and median opening of the datasets of Figure 1, by population; a flight of G and A counts in each population where it has a piece long enough.

**The long lags rest on folded flights.** The declared open flights are far from straight lines: their median opening is 0.32 (Figure 2, Table 4). The longer the lag, the more folded the flights it rests on: the median opening is 0.30 on the dataset up to 2 250 s and 0.23 up to 11 450 s, and the share of flights with an opening below 0.1 grows from 16% to 18% and 26%. A flight that comes back near its takeoff moves away and then back, and its displacement stops growing as the lag nears its duration: the bend of Section 3 is the shape of the flights. In the longest dataset the flights of G are more open than those of S and A (median 0.29 against 0.23 and 0.19), so part of the straighter progress of G at long lags is their shape.

**The threshold.** $H$ has a meaning where the mean squared displacement is a power law, a straight line in log–log, and we want it on flights that are open. A threshold on the opening drops the declared open flights that are not. A strict one, 0.5, keeps only straight flights and raises $H$ by construction; it also keeps 42% of the flights that take off below 1 000 m but only 13% of those above, where relief and airspace do not let a flight go straight (Figure 2, right), so it would select the region as well as the shape. We set the threshold at 0.1: the lowest that drops the flights that come back near their takeoff, it keeps nearly the same share of the two, 88% and 80%. It keeps 55 879 flights, 84% of the declared open ones.

**What we keep.** From here on an *open flight* is a declared open flight with an opening above 0.1.

# 5 Point 1: raw

How do S, A and G compare as they are, on the open flights, with no stratification?

**The MSD.** On dataset 1000, $H_{\text{loc}}(1000\text{ s})$ is 0.915 [0.913, 0.918] in G, 0.897 [0.895, 0.899] in S and 0.894 [0.891, 0.898] in A (Table 5). The gap G − S is +0.018 [+0.015, +0.021], G − A is +0.021 [+0.019, +0.024] and A − S is −0.003 [−0.006, −0.000]. Over the lags (Figure 3) G − S is negative at the short ones, −0.006 [−0.007, −0.004] at 10 s, and positive from 170 s on; A − S is positive at the short lags (+0.005 [+0.003, +0.006] at 10 s) and near 0 at the long ones. On dataset 2250, $H_{\text{loc}}(1840\text{ s})$ is 0.923 in G and 0.906 in S, a gap of +0.017 [+0.014, +0.021]. On the longest dataset G keeps 115 flights only: $H_{\text{loc}}(1000\text{ s})$ is 0.936 [0.920, 0.952] in G and 0.912 in S, and at the longest lags $H_{\text{loc}}$ of S and A falls while that of G does not (Figure 14, Appendix L).

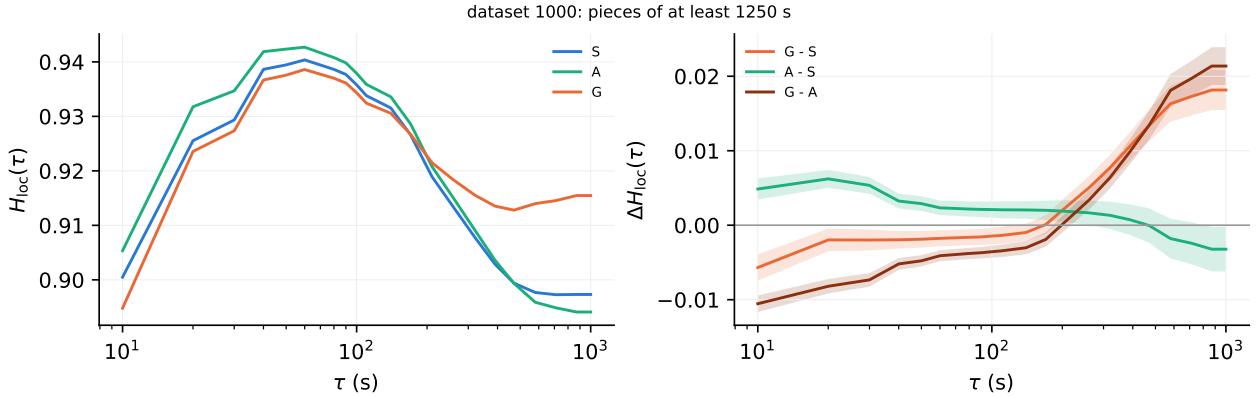



**Figure 3:** Open flights, dataset 1000 (pieces of at least 1 250 s, lags to 1 000 s). Left: $H_{loc}(\tau)$ of S (43 977 flights), A (8 991) and G (11 077), ensemble mean over flights. Right: the differences G − S, A − S and G − A at every lag, with the 90% band of their own draws. MSD($\tau$) and $H_{loc}(\tau)$ with their bands, and the other datasets, in Appendix L.

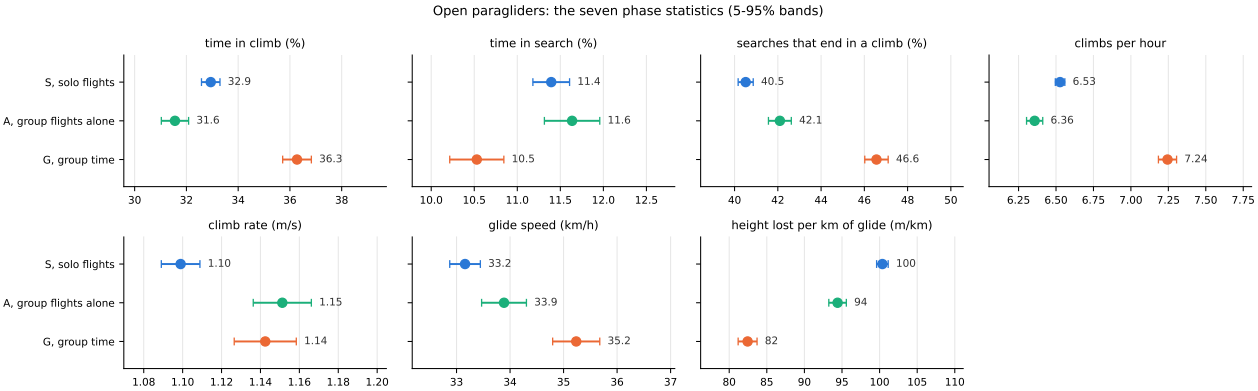



**Figure 4:** Open flights: the seven phase statistics of S (27 665 flights), A (6 397) and G (7 896), on the pieces of dataset 1000; average first, ratio after, unit the flight; 90% bands.

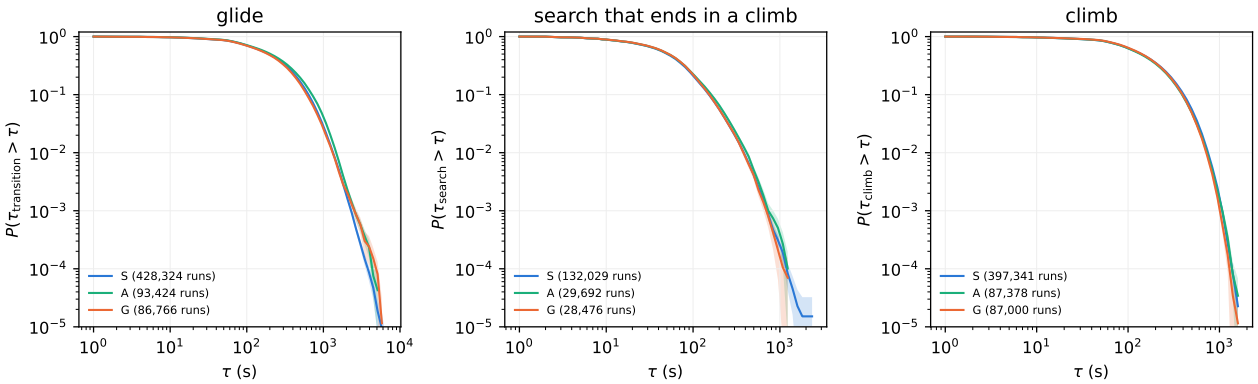



**Figure 5:** Open flights: survival $P(\tau_{phase} > \tau)$ of the duration of glides, of searches that end in a climb and of climbs, on the pieces of dataset 1000, every run weighing 1; runs in the legend; 90% bands at each $t$.

| dataset | | flights | pieces | fitted $H$ | $H_{loc}$(1000 s) | $H_{loc}$(1840 s) |
|---|---|---|---|---|---|---|
| to 1 000 s | S | 43 977 | 46 239 | 0.921 [0.920, 0.922] | 0.897 [0.895, 0.899] | – |
| | A | 8 991 | 13 464 | 0.923 [0.922, 0.925] | 0.894 [0.891, 0.898] | – |
| | G | 11 077 | 14 401 | 0.924 [0.922, 0.925] | 0.915 [0.913, 0.918] | – |
| to 2 250 s | S | 36 975 | 37 865 | 0.916 [0.914, 0.917] | 0.898 [0.896, 0.900] | 0.906 [0.903, 0.909] |
| | A | 6 966 | 8 779 | 0.920 [0.918, 0.922] | 0.897 [0.893, 0.901] | 0.902 [0.897, 0.907] |
| | G | 8 405 | 10 134 | 0.923 [0.922, 0.925] | 0.918 [0.914, 0.921] | 0.923 [0.919, 0.928] |
| to 11 450 s | S | 4 309 | 4 311 | 0.923 [0.920, 0.927] | 0.912 [0.908, 0.916] | 0.922 [0.917, 0.927] |
| | A | 602 | 602 | 0.929 [0.924, 0.935] | 0.917 [0.910, 0.924] | 0.926 [0.919, 0.934] |
| | G | 115 | 115 | 0.943 [0.930, 0.956] | 0.936 [0.920, 0.952] | 0.948 [0.932, 0.965] |

**Table 5:** Open flights, raw: flights and pieces of each population and dataset, fitted $H$ (10 s to the end of the lags), $H_{loc}$(1000 s) and $H_{loc}$(1840 s), with 90% bands. $H_{loc}$(1000 s) on dataset 1000 is measured at the lag 1000 s; $H_{loc}$(1840 s) needs a dataset of at least 2250 s, and its window is cut at 2250 s. Ensemble mean over flights, one weight per flight.

**The phases.** G spends more time in climb than S (36.3% against 32.9%), finds a climb after a search more often (46.6% against 40.5%), makes more climbs per hour (7.24 against 6.53), glides faster (35.2 against 33.2 km/h) and loses less height per km of glide (82 against 100 m/km); its time in search is lower (10.5% against 11.4%). The climb rate is 1.14 against 1.10 m/s. The gaps G − S of the seven statistics have bands that exclude 0: +3.3 [+2.8, +3.8] points of time in climb, +2.1 [+1.7, +2.5] km/h of glide speed, −18 [−19, −17] m/km of height lost. For all statistics but the climb rate A lies nearer to S than to G. The survival functions of the three populations (Figure 5) overlap for climbs and for searches; for glides the tail of S falls below those of A and G at the longest durations, where few runs remain.

**The estimator matters.** The ratio of sums weights a flight by its phase time. It gives 34.2% of time in climb in S against 32.9% for the average first, and 34.2 against 33.2 km/h of glide speed (Appendix D, Table 11).

**What we keep.** Raw, on the open flights, group time progresses straighter than solo flight at the long lags ($H_{\text{loc}}$(1000 s): G − S +0.018 [+0.015, +0.021]) and a little less straight at the short ones, and it gains more height per unit of time: more time in climb, more climbs per hour, a glide that is faster and loses less height. The same flights alone stay close to S. This gap mixes the route, the season, the hour, the wing and the flights themselves, which the next two sections take apart.

# 6 Point 2: the same route, season, hour and wing

Does group time still differ from solo flight when the solo side flies the same route, in the same season and hour, with the same wing?

**Controls.** For each piece of G (a stretch in a group, at least 1 250 s) the controls are solo segments of the pool, the open flights that never had a contact of any length, on another date than the piece. A segment matches when at least 0.8 of the points of the piece, one every 30 s, have the pool track within 500 m, the pool flight flew them in the same direction, and the segment lasts at least as long as the dataset needs (Appendix F). The match table keeps every match with a cover of at least 0.7, the lowest of the robustness lines: 96 462 matches, from 4 518 pool flights; 6 018 pieces of G and 3 851 pieces of A have at least one. A caliper then keeps the matches of the same season (month), the same hour and the same wing.

| | G, dataset to 1000 s | | | G, dataset to 2250 s | | | A, all | |
|---|---|---|---|---|---|---|---|---|
| caliper | beg. | exp. | all | beg. | exp. | all | 1000 | 2250 |
| pieces without a condition | 3 940 | 10 002 | 14 401 | 2 679 | 7 132 | 10 134 | 13 464 | 8 779 |
| ∗ month ±1, hour ±1, same class | 781 | 973 | 1 754 | 311 | 395 | 706 | 988 | 275 |
| month ±1, hour ±1, same level | 823 | 1 511 | 2 334 | 327 | 656 | 983 | 1 298 | 375 |
| month ±2, hour ±2, same class | 1 132 | 1 585 | 2 717 | 513 | 725 | 1 238 | 1 607 | 518 |
| month ±2, hour ±2, same level | 1 181 | 2 251 | 3 432 | 537 | 1 095 | 1 632 | 1 998 | 677 |

**Table 6:** Pieces of G (dataset 1000 and 2250) and of A: all of them (first row), and those with at least one control at cover 0.8 under each caliper, by level (beg. beginners, exp. experienced, wings A and B against C, D and CCC; a wing of another class has no level and leaves); ∗ the chosen caliper. Month and hour are those of the middle of the piece and of the segment.

**The caliper.** The calipers are tried in the order of Table 6, and the chosen one is the first with at least 200 pieces of G per level on dataset 2250: month ±1, hour ±1, the same wing class (311 pieces for beginners, 395 for experienced). It leaves 1 754 pieces of G on dataset 1000, from 1 640 flights and 1 033 pilots, and 988 of A. Isolation (nobody within 10 km) is off in the main result.

**The MSD.** Against its controls, G has $H_{\text{loc}}$(1000 s) of 0.906 and S | G of 0.899: Δ(G − S | G) is +0.007 [−0.000, +0.015] (Table 7, Figure 6), against the raw gap +0.018 of Section 5; its band reaches 0. On dataset 2250, $H_{\text{loc}}$(1840 s) gives +0.018 [+0.004, +0.032], and $H_{\text{loc}}$ at 1000 s, where the window of these longer pieces is not cut, +0.026 [+0.015, +0.037]: both bands exclude 0. Over the lags Δ(G − S | G) is negative at the short ones and positive at the long ones, as the raw gap: the fitted $H$ over 10 to 1000 s is lower in G than in its controls (−0.005 [−0.009, −0.002]), the fitted $H$ over 10 to 2250 s higher (+0.010 [+0.005, +0.015]). The band of Δ(G − S | G) includes 0 for 4 of the 9 quantities of Figure 7.

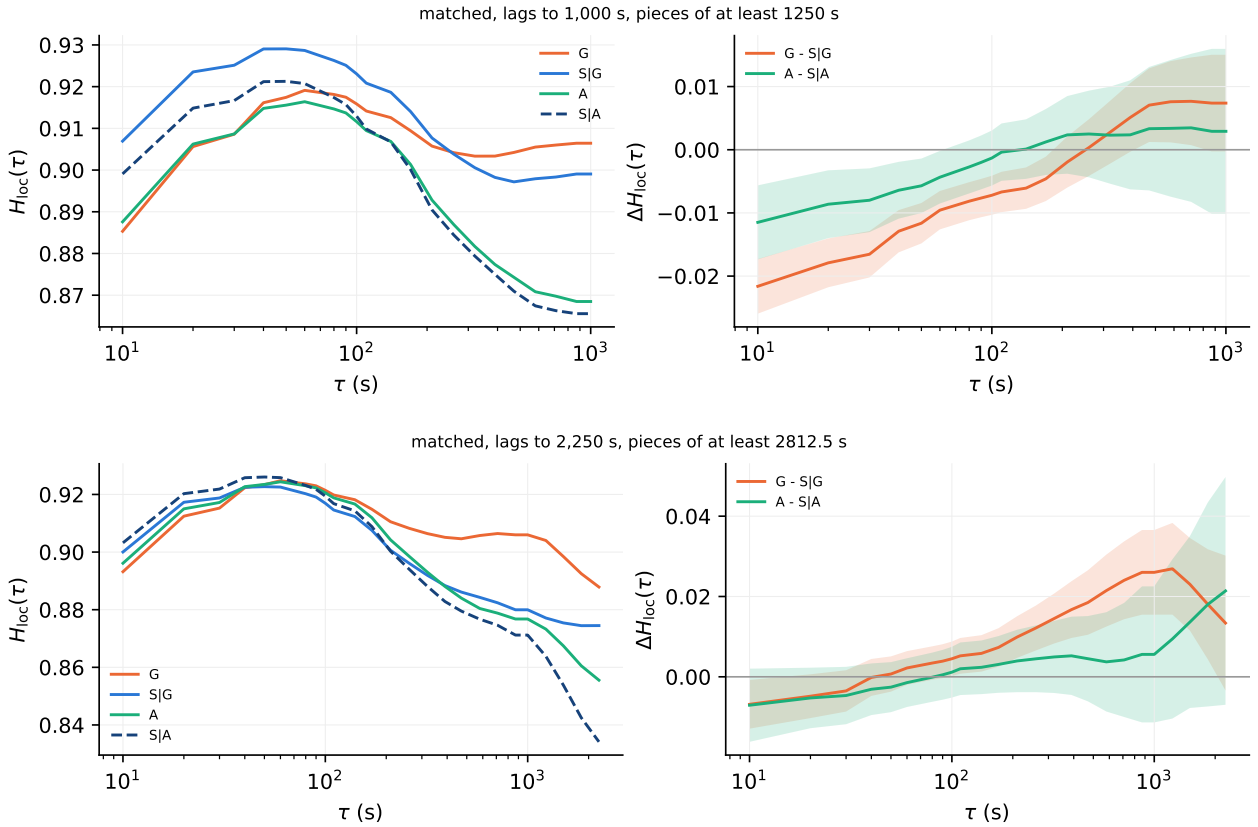



**Figure 6:** Same route, season, hour and wing (caliper month ±1, hour ±1, same class), datasets 1000 (top, pieces of at least 1 250 s) and 2250 (bottom, 2 812.5 s). Left: $H_{\mathrm{loc}}(\tau)$ of G (1 754 pieces on dataset 1000), its controls S | G (3 436 segments), A (988 pieces) and S | A (1 821); each side averaged over units reweighted to the piece (Appendix G). Right: $\Delta(\mathrm{G}-\mathrm{S}\mid\mathrm{G})$ and $\Delta(\mathrm{A}-\mathrm{S}\mid\mathrm{A})$ at every lag, 90% bands. MSD($\tau$) and $H_{\mathrm{loc}}(\tau)$ with their bands in Appendix L.

| | pieces | fitted $H$, 10 to 1000 s | $H_{\mathrm{loc}}$(1000 s) | $H_{\mathrm{loc}}$(1840 s) |
|---|---|---|---|---|
| G | 1 754 | 0.908 [0.904, 0.912] | 0.906 [0.900, 0.913] | 0.893 [0.878, 0.907] |
| S \| G | 3 436 | 0.914 [0.910, 0.918] | 0.899 [0.891, 0.907] | 0.874 [0.858, 0.891] |
| A | 988 | 0.898 [0.892, 0.903] | 0.868 [0.857, 0.880] | 0.861 [0.833, 0.888] |
| S \| A | 1 821 | 0.899 [0.893, 0.905] | 0.866 [0.853, 0.879] | 0.843 [0.808, 0.877] |
| G − S \| G | | −0.005 [−0.009, −0.002] | +0.007 [−0.000, +0.015] | +0.018 [+0.004, +0.032] |
| A − S \| A | | −0.002 [−0.006, +0.003] | +0.003 [−0.010, +0.016] | +0.018 [−0.007, +0.043] |
| difference | | −0.004 [−0.009, +0.001] | +0.004 [−0.010, +0.019] | +0.000 [−0.029, +0.029] |

**Table 7:** Same route, season, hour and wing, level all: pieces (or control segments), fitted $H$ (dataset 1000, 10 to 1000 s), $H_{\mathrm{loc}}$(1000 s) (dataset 1000) and $H_{\mathrm{loc}}$(1840 s) (dataset 2250, 706 pieces of G); differences of the sides with 90% bands. Ensemble mean over units reweighted to the piece.

**Selection.** A against S | A flies the same route alone: same pilots, gliders and days as G, no group. $\Delta(\mathrm{A}-\mathrm{S}\mid\mathrm{A})$ measures selection. Its $H_{\mathrm{loc}}$(1000 s) is +0.003 [−0.010, +0.016] and the band includes 0 for 8 of the 9 quantities; the difference $(\mathrm{G}-\mathrm{S}\mid\mathrm{G})-(\mathrm{A}-\mathrm{S}\mid\mathrm{A})$ is +0.004 [−0.010, +0.019] at $H_{\mathrm{loc}}$(1000 s) and includes 0 for 9 of them.

**The phases.** Time in climb is 29.2% in G and 26.3% in its controls (Δ is +2.9 [+1.3, +4.4]), climbs per hour 7.63 against 6.98 (+0.64 [+0.25, +1.03]), glide speed 29.5 against 30.9 km/h (−1.3 [−2.2, −0.5]) and height lost per km of glide 69 against 64 m/km (+5 [+1, +9]): on the same route group time climbs more often and glides slower and steeper. The bands of the time in search (+1.1 [−0.5, +2.6]), of the searches ending in a climb (+2.8 [−0.4, +5.9]) and of the climb rate (−0.053 [−0.106, +0.001] m/s) include 0. The survival functions of the three phases (Figure 8) lie close to each other; their bands widen in the tails, where few runs remain.

**Robustness.** Over the 9 lines of Figure 9, $\Delta H_{\mathrm{loc}}$(1000 s) stays between +0.007 and +0.012, with a band that includes 0 in 4 of them, and $\Delta H_{\mathrm{loc}}$(1840 s), on the 8 lines whose long series reaches 1840 s, between +0.016 and +0.028, with a band that includes 0 in 0.

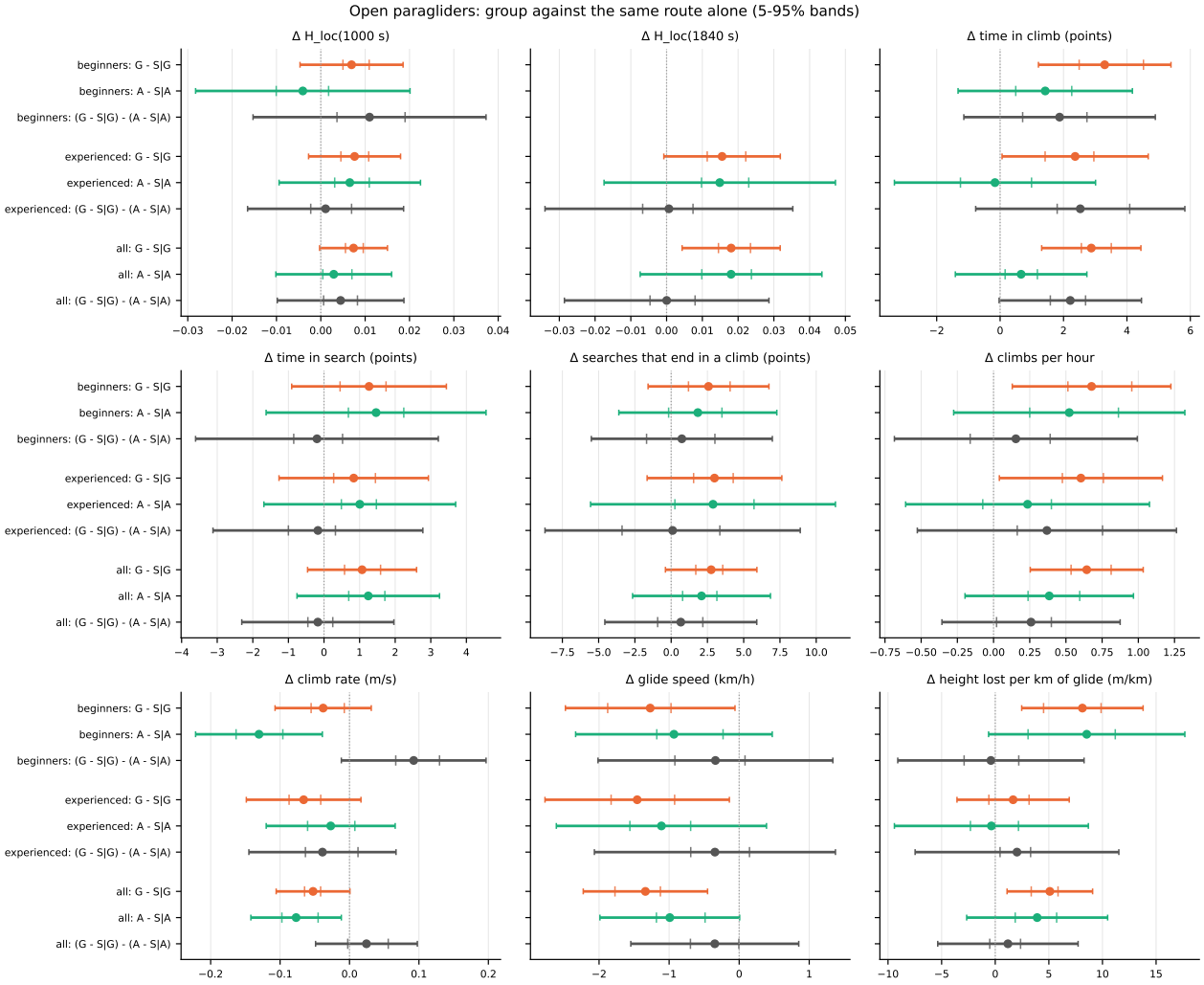



**Figure 7:** Same route, season, hour and wing: $\Delta(\mathrm{G}-\mathrm{S}\mid\mathrm{G})$, $\Delta(\mathrm{A}-\mathrm{S}\mid\mathrm{A})$ and their difference, by level of the wing (beginners, experienced, all). Dot: value; thick bar: 90% band; thin bar: range over the years, leaving one year out. $H_{\mathrm{loc}}(1000\ \mathrm{s})$ and the seven statistics on dataset 1000 (phase statistics on the pieces with phases: 866 pieces of G at level all), $H_{\mathrm{loc}}(1840\ \mathrm{s})$ on the long series, whose lag is the largest at which G and A both keep 200 pieces at the level (2250 s for all, 1 840 s for experienced; beginners stop below 1840 s and have no $H_{\mathrm{loc}}(1840\ \mathrm{s})$); average first, ratio after, unit the piece.

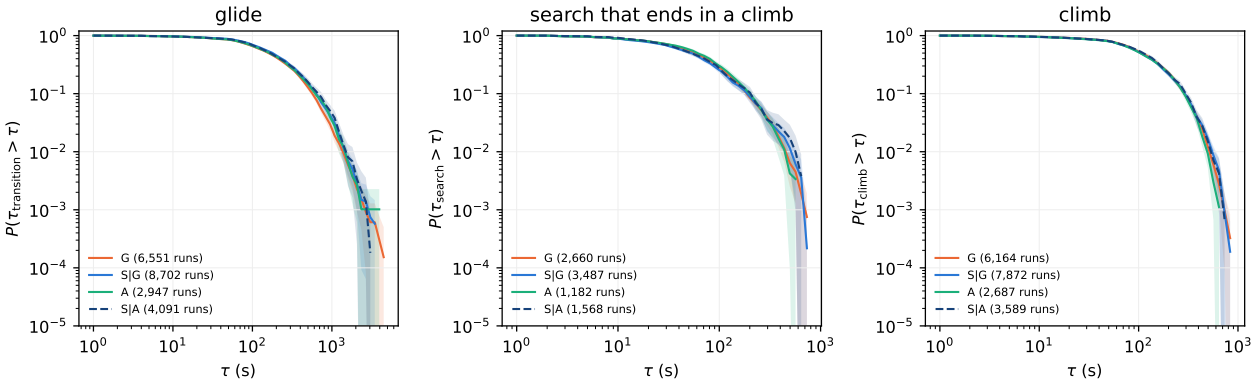



**Figure 8:** Same route, season, hour and wing, level all, dataset 1000: survival of glides, of searches ending in a climb and of climbs for G (runs: 6 164 climbs), its controls (7 872), A (2 687) and S | A (3 589). A run of a piece weighs 1, a run of a control 1/(controls of its piece); 90% bands at each $t$.

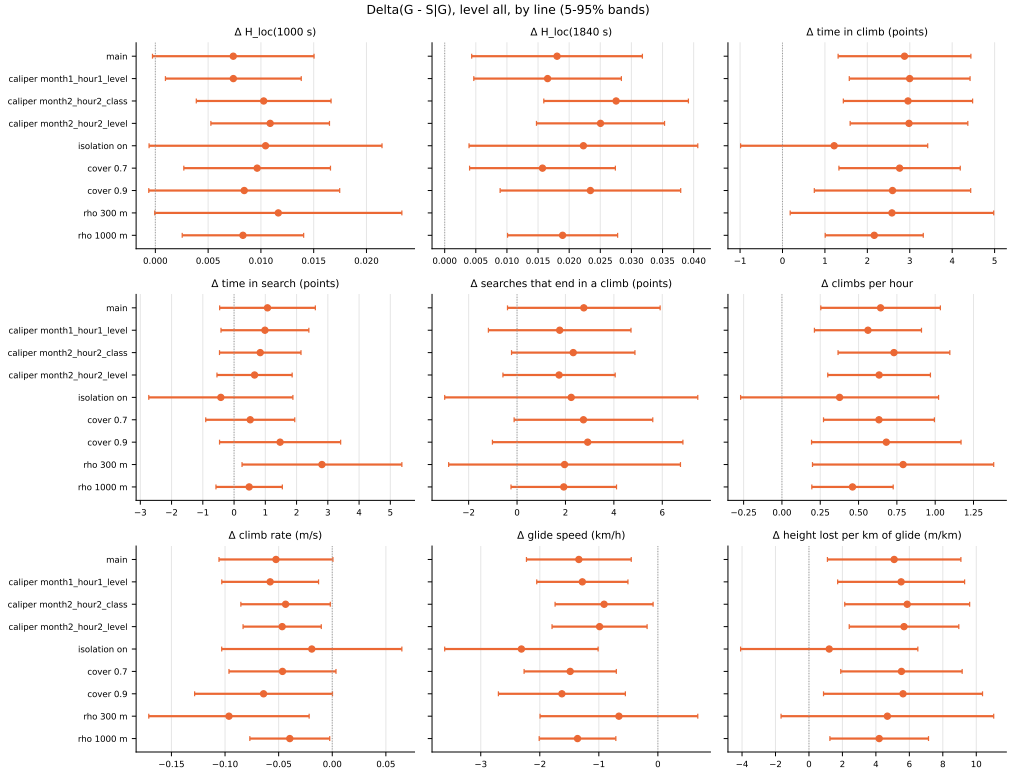



**Figure 9:** Robustness of $\Delta(\mathrm{G}-\mathrm{S} \mid \mathrm{G})$, level all, one row per line of the analysis: the other calipers of Table 6, isolation on (nobody within 10 km: 992 pieces), $\rho = 300$ and 1000 m, cover 0.7 and 0.9. The lines keep between 802 and 3 432 pieces on dataset 1000. 90% bands.

**What we keep.** On the same route, season, hour and wing group time still progresses straighter than solo flight at the long lags: $H_{\mathrm{loc}}(1840\ \mathrm{s})$ $+0.018$ [$+0.004$, $+0.032$], $H_{\mathrm{loc}}$ at 1000 s on dataset 2250 $+0.026$ [$+0.015$, $+0.037$], while on the shorter pieces of dataset 1000 $H_{\mathrm{loc}}(1000\ \mathrm{s})$ gives $+0.007$ [$-0.000$, $+0.015$], against $+0.018$ raw. It is less straight at the short lags, climbs more often and glides slower and steeper than its controls. The selection term $\Delta(\mathrm{A}-\mathrm{S} \mid \mathrm{A})$ is $+0.003$ [$-0.010$, $+0.016$] at $H_{\mathrm{loc}}(1000\ \mathrm{s})$ and its bands include 0 for 8 of the 9 quantities: the pilots who fly in groups, flying alone on the same routes, do not differ from their controls by more than the bands allow.

# 7 Point 3: how much each factor accounts for

How much of the raw gap between group time and solo flight do the route (R), the season and hour (T) and the wing (W) account for, and how much is left to the group?

**Cells.** For each of the 8 subsets $C$ of $F = \{R, T, W\}$, $\Delta(C)$ is G minus the controls that satisfy only the conditions of $C$, on the same 1 754 pieces of G in every cell: those with a control in the cell $F$ (the caliper of Section 6). $\Delta(\varnothing)$ compares the pieces with solo time of any route, season, hour and wing; $\Delta(F)$ is the result of Section 6. In the cells with R the controls are the matches that satisfy $C$. In the cells without R a control is a window of a solo piece of the pool, as long as the piece, drawn at random (at most 50 per piece) among the pool pieces that satisfy $C$ and are long enough: with T, the pool piece in the month caliper and the window placed where its middle is within the hour caliper; with W, the same wing class (Appendix F). The Shapley value $\varphi_k$ is the part of $\Delta(\varnothing)$ that comes from G and its controls differing in $k$ (Appendix J):

$$\Delta(\varnothing) = \Delta(F) + \varphi_R + \varphi_T + \varphi_W \qquad \text{(exact, on every draw).}$$

**The parts.** For the time in climb, $\Delta(\varnothing)$ is $-9.6$ [$-10.6$, $-8.6$]: the pieces of G climb less than the controls of any route. The route accounts for $-11.7$ [$-12.9$, $-10.5$], season and hour for $-0.1$, the wing for $-0.7$, and $\Delta(F)$ is $+2.9$ [$+1.3$, $+4.4$]. The share of $\varphi_R$ in $\Delta(\varnothing)$ is 122% and that of $\Delta(F)$ is $-30$%: they pass 100% and 0 because $\Delta(F)$ has the opposite sign. The glide statistics follow the same pattern, $\varphi_R$ carrying most of $\Delta(\varnothing)$: for the glide speed $\Delta(\varnothing)$ is $-5.7$, $\varphi_R$ $-3.6$ and $\Delta(F)$ $-1.3$; for the height lost per km of glide $\Delta(\varnothing)$ is $-27$, $\varphi_R$ $-34$ and $\Delta(F)$ $+5$. At $H_{\mathrm{loc}}(1840\ \mathrm{s})$, $\Delta(\varnothing)$ is $-0.032$ [$-0.043$, $-0.021$] and $\varphi_R$ is $-0.048$ [$-0.059$, $-0.036$]. At $H_{\mathrm{loc}}(1000\ \mathrm{s})$, $\Delta(\varnothing)$ is $-0.002$ [$-0.007$, $+0.004$], near 0 with a band that includes it, so no share is quoted; the

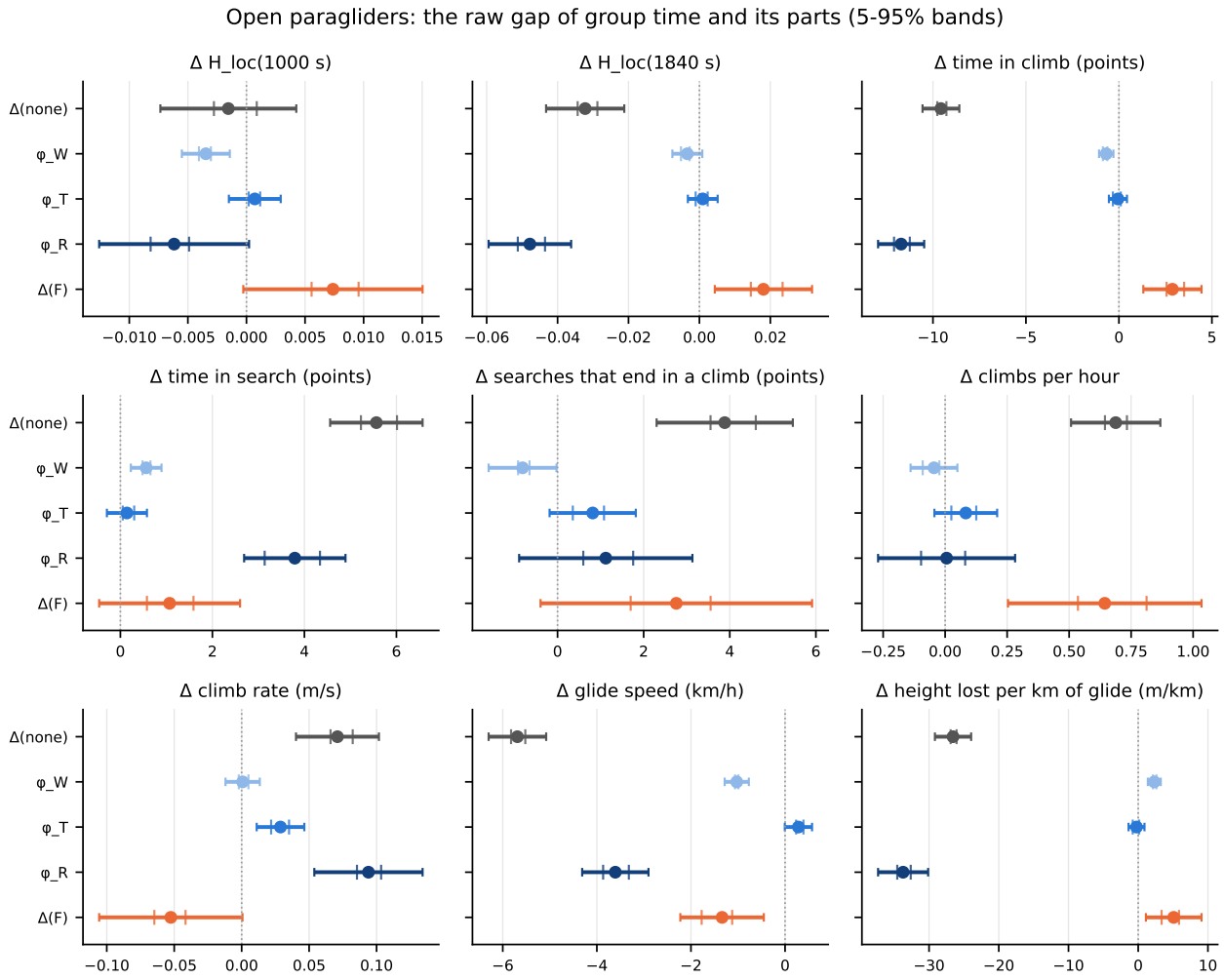



**Figure 10:** The raw gap and its parts, level all, on 1 754 pieces of G (866 with phase time for the seven statistics); $H_{\mathrm{loc}}(1000\ \mathrm{s})$ on dataset 1000, $H_{\mathrm{loc}}(1840\ \mathrm{s})$ on dataset 2250 (706 pieces). From the top: $\Delta(\varnothing)$, $\varphi_W$, $\varphi_T$, $\varphi_R$, $\Delta(F)$. Thick bar: 90% band, the same draws in every cell; thin: range over the years. Average first, ratio after, unit the piece.

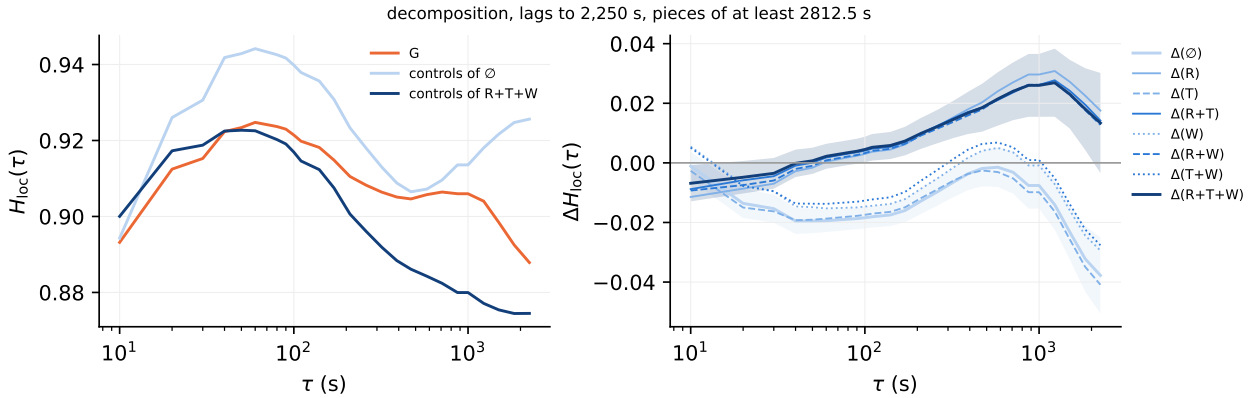



**Figure 11:** Dataset 2250 (706 pieces of at least 2 812.5 s). Left: $H_{\mathrm{loc}}(\tau)$ of G and of the controls of the cells $\varnothing$ and $F$. Right: $\Delta(C)$ of the 8 cells at every lag, the darker the more conditions, with the 90% bands of $\Delta(\varnothing)$ and $\Delta(F)$. Dataset 1000, and $\mathrm{MSD}(\tau)$ and $H_{\mathrm{loc}}(\tau)$ of every cell with their bands, in Appendix L.

| | G | raw gap | $\Delta(\emptyset)$ | | $\varphi_R$ | $\varphi_T$ | $\varphi_W$ | $\Delta(F)$ | |
|---|---|---|---|---|---|---|---|---|---|
| | pieces | point 1 | controls | $\Delta$ | | | | controls | $\Delta$ |
| $H_{\text{loc}}(1000\text{ s})$ | 0.906 | +0.018 | 0.908 | −0.002 | −0.006 | +0.001 | −0.003 | 0.899 | +0.007 |
| $H_{\text{loc}}(1840\text{ s})$ | 0.893 | +0.017 | 0.925 | −0.032 | −0.048 | +0.001 | −0.003 | 0.874 | +0.018 |
| time in climb (%) | 29.2 | +3.3 | 38.8 | −9.6 | −11.7 | −0.1 | −0.7 | 26.3 | +2.9 |
| time in search (%) | 15.3 | −0.9 | 9.7 | +5.6 | +3.8 | +0.1 | +0.6 | 14.2 | +1.1 |
| searches ending in a climb (%) | 49.5 | +6.0 | 45.6 | +3.9 | +1.1 | +0.8 | −0.8 | 46.7 | +2.8 |
| climbs per hour | 7.63 | +0.72 | 6.94 | +0.69 | +0.01 | +0.08 | −0.05 | 6.98 | +0.64 |
| climb rate (m/s) | 1.16 | +0.044 | 1.09 | +0.071 | +0.094 | +0.029 | +0.001 | 1.22 | −0.053 |
| glide speed (km/h) | 29.5 | +2.1 | 35.2 | −5.7 | −3.6 | +0.3 | −1.0 | 30.9 | −1.3 |
| height lost per km of glide (m/km) | 69 | −18 | 95 | −27 | −34 | −0 | +2 | 64 | +5 |

**Table 8:** Decomposition, level all: value of G on its 1 754 pieces (706 for $H_{\text{loc}}(1840\text{ s})$, 866 for the phase statistics), the raw gap G − S of Section 5, the controls and Δ of the cells ∅ and $F$, and $\varphi_R$, $\varphi_T$, $\varphi_W$. Dataset 1000 ($H_{\text{loc}}(1000\text{ s})$, statistics), 2250 ($H_{\text{loc}}(1840\text{ s})$); average first, ratio after, unit the piece. The eight cells in Appendix L.

$\varphi$ are −0.006 [−0.013, +0.000] (R), +0.001 (T) and −0.003 [−0.006, −0.001] (W). Over the lags (Figure 11) $\Delta(\emptyset)$ and the cells without R lie below 0 at the long lags, and the cells with R near or above it.

**Why $\Delta(\emptyset)$ is not the raw gap.** The raw gap of Section 5 is +3.3 in time in climb, +2.1 km/h in glide speed, −18 m/km in height lost, +0.018 in $H_{\text{loc}}(1000\text{ s})$ and +0.017 in $H_{\text{loc}}(1840\text{ s})$; $\Delta(\emptyset)$ is −9.6, −5.7, −27, −0.002 and −0.032. Three things differ.

(a) *The pieces of* G. Only the pieces with a route match in the cell $F$ enter, 1 754 of the 14 401 pieces of dataset 1000. Their $H_{\text{loc}}(1000\text{ s})$, time in climb and glide speed are lower than those of all of G: $H_{\text{loc}}(1000\text{ s})$ is 0.906 on them and 0.915 on all of G; time in climb is 29.2% against 36.3%, glide speed 29.5 against 35.2 km/h.

(b) *The controls without R.* Their windows are as long as the piece and come from solo pieces at least that long, so from long flights. These climb more and glide faster than the solo flights of Section 5: time in climb 38.8% in the controls of ∅ against 32.9% in S (34.2% with the ratio of sums, in which a flight weighs by its phase time), glide speed 35.2 against 33.2 km/h. $H_{\text{loc}}(1000\text{ s})$ of S rises with the length of the pieces, mostly beyond 2250 s: 0.897 on dataset 1000, 0.898 on 2250 and 0.912 on the longest; it is 0.908 in the controls of ∅.

(c) *The unit.* The unit is the piece, not the flight: the 1 754 pieces come from 1 640 flights, and the controls of a piece are averaged before the sum over pieces.

The controls with R have 26.9% of time in climb: a route fixes the flight from the start to the end of the piece, in the same direction, so $\varphi_R$ carries that constraint together with the route.

**What we keep.** The raw gap does not carry through. Against controls of any route, group pieces climb less (−9.6) and glide slower (−5.7 km/h), and at $H_{\text{loc}}(1840\text{ s})$ they are less straight (−0.032); the route accounts for the largest part of that (−11.7, −3.6 km/h, −0.048), season and hour for almost none, the wing for a small part (−1.0 km/h of glide speed). What remains after the three factors is $\Delta(F)$ of Section 6, of the opposite sign for time in climb and $H_{\text{loc}}(1840\text{ s})$: on the same route, season, hour and wing, group time climbs more (+2.9 [+1.3, +4.4]) and is straighter at the long lags (+0.018 [+0.004, +0.032]).

# 8 Point 4: before, during, after

Does the same flight move differently inside its group time than just before and after it?

**Windows.** For each stretch of group time of a flight, four windows of 2 400 s of the flight clock: *pre*, the 2 400 s before the stretch, if a solo stretch of the same flight at least that long precedes it; *start* and *end*, its first and last 2 400 s, if it lasts that long; *post*, the 2 400 s after it, if a solo stretch at least that long follows. A window is the unit. The MSD is measured on the 24 lags of the grid up to 1 840 s, the largest within $0.8 \times 2\,400$ s, with 56 origins at the longest. The windows of one stretch are one event (1 595 events with the four windows). The controls are two consecutive windows of 2 400 s drawn at random in the solo time of the flights of A (5 205 events) and in solo flights (26 175 events).

**The MSD.** $H_{\text{loc}}(1000\text{ s})$ is measured, as in the other points, with its window cut at 1000 s. On the events with the four windows it is 0.924 (start) and 0.926 (end) inside the group, against 0.899 before and 0.899 after (Figure 12). The entry difference $D_{\text{in}} = \text{start} - \text{pre}$ is +0.025 [+0.017, +0.034] and the exit difference $D_{\text{out}} = \text{end} - \text{post}$ is +0.027 [+0.019, +0.035]; after minus before, on the 1 818 events with both, is +0.002

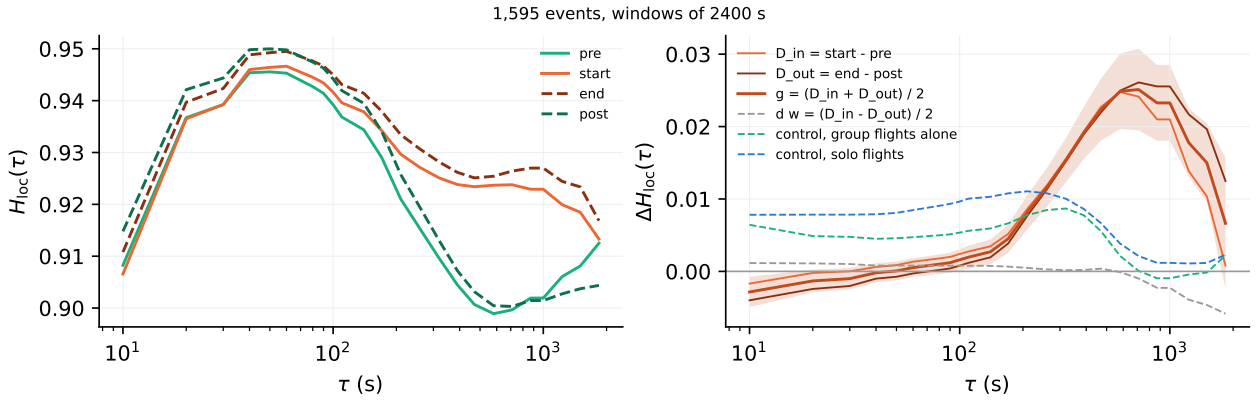



**Figure 12:** The 1 595 events with the four windows (1 595 windows each), lags to 1 840 s. Left: $H_{loc}(\tau)$ of pre, start, end and post, ensemble mean over windows. Right: $D_{in}$, $D_{out}$, $g$ (with its 90% band), $d_w$ and the two control pairs (later minus earlier window), at every lag. $H_{loc}(1000\text{ s})$, with its window cut at 1000 s as on dataset 1000: pre 0.899, start 0.924, end 0.926, post 0.899. MSD$(\tau)$ and $H_{loc}(\tau)$ with their bands in Appendix L.

$[-0.006, +0.009]$. The control pairs give $-0.000$ $[-0.005, +0.004]$ (same flights alone) and $+0.002$ $[-0.000, +0.004]$ (solo flights).

**The drift-free effect.** If the quantity drifts linearly in time (the day advances, the pilot tires) the entry difference is the group effect plus the drift over one window, and the exit difference the group effect minus it. So

$$g = \frac{D_{in} + D_{out}}{2} \quad \text{(the group effect)}, \qquad d_w = \frac{D_{in} - D_{out}}{2} \quad \text{(the drift over one window)},$$

on the events that have the four windows (1 595 for the MSD, 1 150 with phase time; Appendix K). At $H_{loc}(1000\text{ s})$, $g$ is $+0.026$ $[+0.020, +0.032]$ and $d_w$ is $-0.001$ $[-0.006, +0.004]$.

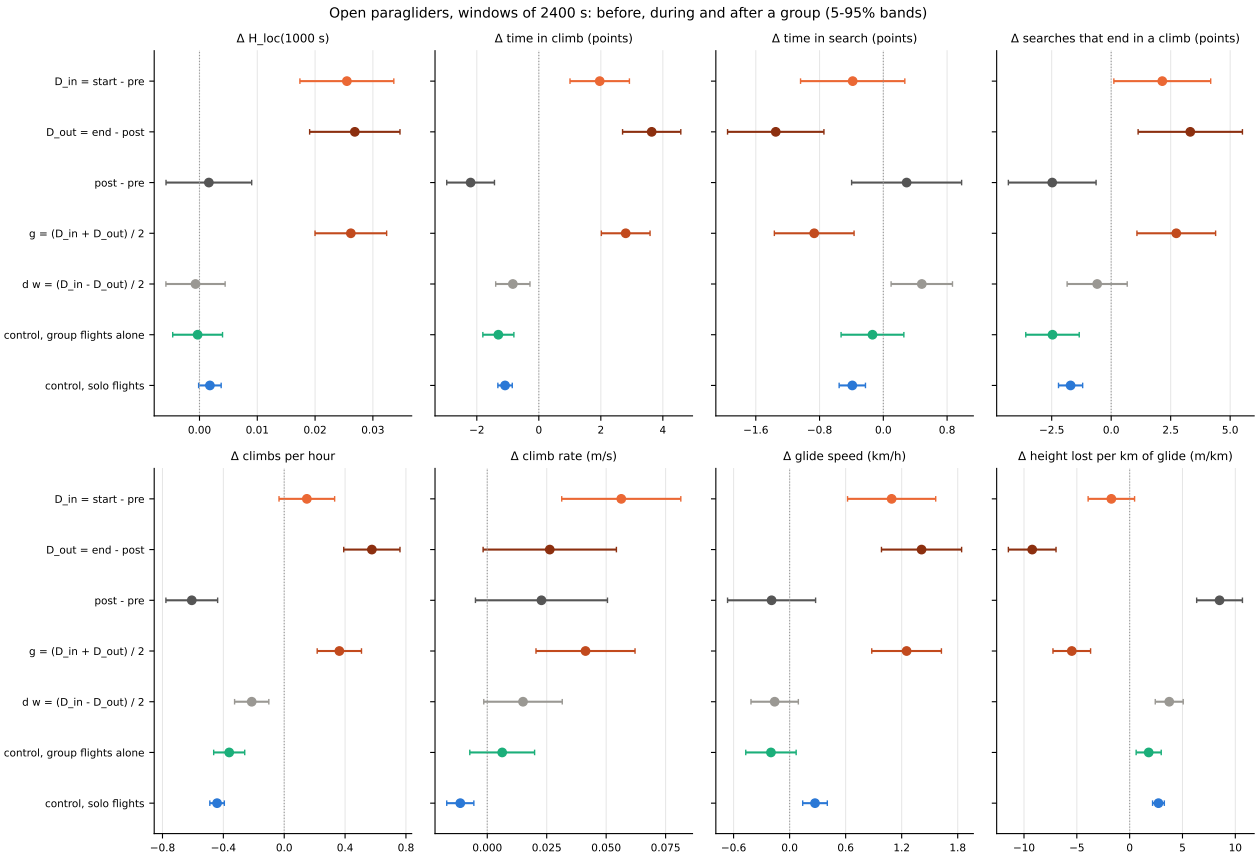



**Figure 13:** Before, during and after: $\Delta H_{loc}(1000\text{ s})$ and the seven statistics for $D_{in}$, $D_{out}$, post minus pre, $g$, $d_w$ and the two control pairs; windows of 2 400 s, $H_{loc}(1000\text{ s})$ on the MSD of the windows (1 595 events for $D_{in}$, $D_{out}$, $g$ and $d_w$), statistics on 1 150 events with phase time; average first, ratio after, unit the window; 90% bands.

**The phases.** Time in climb is higher inside the group than before it ($D_{\text{in}}$ +2.0) and than after it ($D_{\text{out}}$ +3.6), and $g$ is +2.8 [+2.0, +3.6] points; the control pairs give −1.1 (solo flights) and −1.3 (same flights alone). For the glide speed $g$ is +1.3 [+0.9, +1.6] km/h and for the height lost per km of glide −5 [−7, −4] m/km; for the climbs per hour it is +0.36 [+0.22, +0.51].

| | $g$, before and after | $\Delta(\text{G} - \text{S} \mid \text{G})$, same route | $g - \Delta$ | same sign |
|---|---|---|---|---|
| $H_{\text{loc}}(1000\text{ s})$ | +0.026 [+0.020, +0.032] | +0.007 [−0.000, +0.015] | +0.019 | yes |
| time in climb (%) | +2.8 [+2.0, +3.6] | +2.9 [+1.3, +4.4] | −0.1 | yes |
| time in search (%) | −0.9 [−1.4, −0.4] | +1.1 [−0.5, +2.6] | −1.9 | no |
| searches ending in a climb (%) | +2.7 [+1.1, +4.4] | +2.8 [−0.4, +5.9] | −0.0 | yes |
| climbs per hour | +0.36 [+0.22, +0.51] | +0.64 [+0.25, +1.03] | −0.28 | yes |
| climb rate (m/s) | +0.041 [+0.020, +0.062] | −0.053 [−0.106, +0.001] | +0.094 | no |
| glide speed (km/h) | +1.3 [+0.9, +1.6] | −1.3 [−2.2, −0.5] | +2.6 | no |
| height lost per km of glide (m/km) | −5 [−7, −4] | +5 [+1, +9] | −11 | no |

**Table 9:** The drift-free effect $g$ of this section (windows of 2 400 s, average first, unit the window) beside $\Delta(\text{G} - \text{S} \mid \text{G})$ of Section 6 (pieces, dataset 1000, same route, season, hour and wing, level all), at the same lag, with 90% bands; $g - \Delta$ and whether the two have the same sign.

**Against the same route.** At $H_{\text{loc}}(1000\text{ s})$, $g$ (+0.026 [+0.020, +0.032]) lies above the band of $\Delta(\text{G} - \text{S} \mid \text{G})$ on dataset 1000 (+0.007 [−0.000, +0.015]), +0.019 above its point value. The windows last 2 400 s, while the pieces of dataset 1000 can be as short as 1 250 s; on the longer pieces of dataset 2250, $\Delta(\text{G} - \text{S} \mid \text{G})$ at 1000 s is +0.026 [+0.015, +0.037], close to $g$. Time in climb, searches ending in a climb and climbs per hour have the same sign in the two comparisons (+2.8 against +2.9, +2.7 against +2.8, +0.36 against +0.64). Time in search (−0.9 against +1.1), climb rate (+0.041 against −0.053), glide speed (+1.3 against −1.3) and height lost per km of glide (−5 against +5) have opposite signs (Table 9); for the glide speed and the height lost both bands exclude 0. Inside its group a pilot glides faster and flatter than just before and after, while group time glides slower and steeper than solo flights on the same route.

**What we keep.** The same flight is straighter inside its group than just before and just after it ($D_{\text{in}}$ +0.025, $D_{\text{out}}$ +0.027) while the control pairs stay near 0, so $g$ is +0.026 [+0.020, +0.032], with $d_w$ −0.001 [−0.006, +0.004]. $g$ is close to the same-route gap at 1000 s on the longer pieces of dataset 2250 and above that on dataset 1000. The two comparisons agree in sign for time in climb, searches ending in a climb and climbs per hour, and disagree for time in search, climb rate, glide speed and height lost per km of glide.

# 9 What we keep

| | raw G − S | matched $\Delta(\text{G} - \text{S} \mid \text{G})$ | selection $\Delta(\text{A} - \text{S} \mid \text{A})$ | before and after $g$ |
|---|---|---|---|---|
| $H_{\text{loc}}(1000\text{ s})$ | +0.018 [+0.015, +0.021] | +0.007 [−0.000, +0.015]° | +0.003 [−0.010, +0.016]° | +0.026 [+0.020, +0.032] |
| $H_{\text{loc}}(1840\text{ s})$ | +0.017 [+0.014, +0.021] | +0.018 [+0.004, +0.032] | +0.018 [−0.007, +0.043]° | – |
| time in climb (%) | +3.3 [+2.8, +3.8] | +2.9 [+1.3, +4.4] | +0.7 [−1.4, +2.7]° | +2.8 [+2.0, +3.6] |
| time in search (%) | −0.9 [−1.2, −0.6] | +1.1 [−0.5, +2.6]° | +1.2 [−0.8, +3.2]° | −0.9 [−1.4, −0.4] |
| searches ending in a climb (%) | +6.0 [+5.5, +6.6] | +2.8 [−0.4, +5.9]° | +2.1 [−2.7, +6.8]° | +2.7 [+1.1, +4.4] |
| climbs per hour | +0.72 [+0.65, +0.78] | +0.64 [+0.25, +1.03] | +0.38 [−0.20, +0.97]° | +0.36 [+0.22, +0.51] |
| climb rate (m/s) | +0.044 [+0.030, +0.057] | −0.053 [−0.106, +0.001]° | −0.077 [−0.142, −0.012] | +0.041 [+0.020, +0.062] |
| glide speed (km/h) | +2.1 [+1.7, +2.5] | −1.3 [−2.2, −0.5] | −1.0 [−2.0, +0.0]° | +1.3 [+0.9, +1.6] |
| height lost per km of glide (m/km) | −18 [−19, −17] | +5 [+1, +9] | +4 [−3, +10]° | −5 [−7, −4] |

**Table 10:** The four comparisons side by side, with 90% bands. Raw: G − S on the flights of Section 5 (dataset 1000 for $H_{\text{loc}}(1000\text{ s})$, 2250 for $H_{\text{loc}}(1840\text{ s})$, average first, unit the flight); same route, selection: pieces, level all; before and after: $g$ on windows. °: the band includes 0. $g$ has no value at $H_{\text{loc}}(1840\text{ s})$: the windows are too short.

- Many flights declared open fold back on themselves, and the longest lags rest on those. An opening above 0.1 drops the ones that come back near their takeoff and keeps nearly the same share of lowland and mountain flights.
- **Raw.** On the open flights group time progresses straighter than solo flight at the long lags: $H_{\text{loc}}(1000\text{ s})$ 0.915 in G, 0.897 in S, 0.894 in A, a gap G − S of +0.018 [+0.015, +0.021]; at the short lags it is a little less straight. It has more time in climb (36.3% against 32.9%) and a faster glide (35.2 against 33.2 km/h); the bands of the gaps exclude 0 for 7 of the 7 statistics.
- **Same route, season, hour and wing.** The gap of $H_{\text{loc}}(1000\text{ s})$ is +0.007 [−0.000, +0.015], a band that reaches 0; that of $H_{\text{loc}}(1840\text{ s})$ is +0.018 [+0.004, +0.032], and $H_{\text{loc}}$ at 1000 s on the longer pieces +0.026 [+0.015, +0.037]. Group time climbs more (+2.9 points of time in climb) and glides slower and steeper

than its controls. The bands include 0 for 4 of the 9 quantities, and those of the selection term $\Delta(\mathrm{A}-\mathrm{S}\mid\mathrm{A})$ for 8 of them: on the same route the pilots who fly in groups do not differ from their controls when alone, and their group time does at the long lags.

- **Decomposition.** The route carries the largest part of $\Delta(\varnothing)$ for time in climb, glide speed, height lost per km of glide and $H_{\mathrm{loc}}(1840\ \mathrm{s})$; season and hour carry almost none, and the wing a small part. $\Delta(\varnothing)$ differs from the raw gap because of the pieces, the controls without a route and the unit. At $H_{\mathrm{loc}}(1000\ \mathrm{s})$ it is near 0 and no share is given.
- **Before and after.** The drift-free effect $g$ of $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is $+0.026$ [$+0.020$, $+0.032$], above the band of the same-route gap on dataset 1000 and close to the same-route gap at 1000 s on the longer pieces ($+0.026$). Time in climb, searches ending in a climb and climbs per hour agree in sign with the same-route gap. Time in search, climb rate, glide speed and height lost per km of glide have the opposite sign; for the glide speed and the height lost both bands exclude 0. We give no reason for it.

Where a band includes 0 (Table 10, °) the comparison does not tell the sign.

# A TA-MSD and ensemble MSD

For a flight $f$ and a population, the pieces on the 10-s grid give the positions $\mathbf{r}_i$ ($i$ the grid index, $\tau = k\cdot 10$ s). The time-averaged MSD pools the origins of all its pieces,

$$\overline{\delta_f^2}(k) = \frac{1}{N_f(k)} \sum_{\text{pieces}} \sum_i \left\|\mathbf{r}_{i+k} - \mathbf{r}_i\right\|^2,$$

$N_f(k)$ the number of pairs $(i, i+k)$ inside a piece. The ensemble MSD is the mean over the flights with a piece in the population, one weight per flight: $\mathrm{MSD}(\tau) = \frac{1}{F}\sum_f \overline{\delta_f^2}(\tau)$. The sums are computed with the fast Fourier transform and equal the direct sums.

*Example.* Positions on a line $\mathbf{r} = 0, 3, 4, 10$ (a single piece): at the lag of one step, MSD $= 15.33$, from the squared steps; at two steps, MSD $= 32.5$, from $(\mathbf{r}_2 - \mathbf{r}_0)^2$ and $(\mathbf{r}_3 - \mathbf{r}_1)^2$.

# B Fitted $H$ and $H_{\mathrm{loc}}$

$H$ is half the slope of $\log_{10}\mathrm{MSD}$ against $\log_{10}\tau$, least squares with one weight per lag, over all the lags of a curve (10 s to the end of its dataset): $H = \frac{1}{2}\,\text{slope}$. $H_{\mathrm{loc}}(\tau)$ is the same fit over the lags $\lambda$ with $|\log_{10}\lambda - \log_{10}\tau| \le 0.25$; when fewer than 3 lags lie there, the 3 nearest. A straight line gives 1, a random walk $\frac{1}{2}$.

*Example.* With the two lags of one and two steps and the MSD above, the slope of the log–log line is $\log_2(32.5/15.33)$, so $H = 0.54$.

*The window cut at the edge.* $H_{\mathrm{loc}}(1840\ \mathrm{s})$ uses the lags within 0.25 decades of 1840 s, from 1 035 to $1840 \times 10^{0.25}$ s; dataset 2250 stops at 2250 s, so the window is cut on the upper side and uses the lags from 1 035 to 2250 s only. Its band includes this truncation, since every draw is cut the same way. The window of $H_{\mathrm{loc}}(1000\ \mathrm{s})$ on dataset 1000 is cut in the same way at 1000 s.

# C Datasets and the $\tau/0.8$ rule

A piece of $n$ grid points lasts $\ell = 10\,(n-1)$ s and gives, at lag $\tau$, $n - \tau/10 = (\ell - \tau)/10 + 1$ pairs of origins. Requiring $\ell \ge \tau/0.8$ gives origins over at least $0.2\,\ell$ of the piece, the same share of it at the longest lag the piece serves, whatever the piece. A curve to the lag $T$ would mix populations if each lag kept the pieces long enough for itself (the long lags would rest on fewer, longer pieces). A dataset to $T$ keeps the pieces of at least $T/0.8$ s and measures them at every lag up to $T$, so the same pieces carry the whole curve. Datasets 1000 and 2250 have $T = 1\,000$ and $2\,250$ s (pieces of at least 1 250 and 2 812.5 s); the longest has $T = 11\,450$ s (14 312.5 s), the largest lag of the grid at which each population keeps 100 flights. The windows of Section 8 have 2 400 s, so $T$ is the largest lag of the grid up to $0.8 \times 2\,400$ s, 1 840 s.

# D The phase statistics: average first against the ratio of sums

For a unit $u$ (flight, piece, control segment or window) the terms $n_u$, $d_u$ are the sums of the numerator and of the denominator over the runs whose middle lies in $u$, and $T_u$ is the sum of the durations of its runs. The

ratio of sums is $\sum_u n_u / \sum_u d_u$: a unit weighs by its $d_u$, hence by its phase time. The estimator used here is

$$\text{stat} = \frac{\sum_u w_u \, n_u / T_u}{\sum_u w_u \, d_u / T_u}.$$

For the time shares and the climbs per hour the denominator $d_u / T_u$ is the same constant for every unit, so the statistic is the mean over units of a share or a rate. For the others it is a ratio of means of rates per unit of phase time.

*Example.* Three units, searches that end in a climb out of all searches, $(n, d, T) = (1, 10, 100)$, $(5, 20, 400)$, $(2, 30, 500)$. The ratio of sums is $8/60 = 13.3\%$; the mean of the unit shares $n/d$ is 13.9%; the estimator is $\sum n/T \,/\, \sum d/T = 12.6\%$.

| | S | A | G |
|---|---|---|---|
| flights | 27 665 | 6 397 | 7 896 |
| phase time (h) | 60 882 | 13 898 | 12 338 |
| time in climb (%) | 32.9 / 34.2 | 31.6 / 32.0 | 36.3 / 36.4 |
| time in search (%) | 11.4 / 10.3 | 11.6 / 10.3 | 10.5 / 9.9 |
| searches ending in a climb (%) | 40.5 / 43.1 | 42.1 / 44.8 | 46.6 / 47.2 |
| climbs per hour | 6.53 / 6.59 | 6.36 / 6.35 | 7.24 / 7.12 |
| climb rate (m/s) | 1.10 / 1.16 | 1.15 / 1.23 | 1.14 / 1.16 |
| glide speed (km/h) | 33.2 / 34.2 | 33.9 / 34.9 | 35.2 / 36.1 |
| height lost per km of glide (m/km) | 100 / 96 | 94 / 89 | 82 / 82 |

**Table 11:** Open flights, raw, pieces of dataset 1000, unit the flight: each statistic as *average first, ratio after / ratio of sums*, for S, A and G, with the flights and the phase time (h) they rest on.

# E Survival function and its weighted version

For the durations $x_j$ of the counted runs with weights $w_j$, $P(\tau_{\text{phase}} > t) = \sum_j w_j [x_j > t] \big/ \sum_j w_j$, on a logarithmic grid of $t$. The duration is the number of fixes of the run (1 Hz), trimmed at the 0.0001 and 99.9999 percentiles; a search counts only if the next run of the flight is a climb. A run cut by the start or the end of its flight or of its phase segment counts as a complete run with the duration it has, so the survival is a little low at the longest durations; the edges of the units do not cut runs, since a run belongs whole to the unit that holds its middle. In Section 5 every run weighs 1. In Section 6 a run of a piece weighs 1 and a run of a control segment weighs $1/m_g$, with $m_g$ the number of controls of its piece $g$, so that the control side has the composition of the pieces. A draw of the bootstrap multiplies the weight of a run by the weight of its unit; the band holds at each $t$ separately (Appendix H).

*Example.* Durations 10, 30, 30, 100 s with weight 1: $P(\tau > 20) = 0.75$ and $P(\tau > 30) = 0.25$.

# F Route matching and calipers

A query piece (of G or A) is a path on the 10-s grid. A segment of a pool piece (an open flight with no contact of any length, on another date) matches it when

- *coverage*: at least 0.8 of the points of the query, one every 30 s, have a point of the pool track within $\rho = 500$ m;
- *direction and segment*: the points of the pool track nearest to the first and to the last point of the query are within $\rho$ of them, and the first comes before the last: the pool flight flew the route the same way; the segment is the stretch between those points;
- the segment lasts at least as long as the dataset needs.

A query keeps one match per pool flight, the most covered. A caliper is then a filter on the match table: the month of the middles within $m$ (circular), the local hour of the middles within $h$ (circular), and the same wing class (A, B, C, D, CCC of the catalogue; a flight with no class is dropped) or the same level (beginners A and B, experienced C, D and CCC). The calipers are tried in the order of Table 6; the chosen one is the first with at least 200 pieces per level on dataset 2250. The robustness lines change one rule at a time: another caliper, isolation (at every point of the segment, no other flight within 10 km at the same minute), $\rho$ of 300 or 1000 m, cover 0.7 or 0.9.

*Cells without R (Section 7).* The control is a window of a pool piece, with as many grid points as the piece of G, on another date. With T: the pool piece is within the month caliper and the window is placed at random among the positions whose middle is within the hour caliper (local time across midnight); without T: at

random. With W: the pool piece has the wing class of the piece. At most 50 windows per piece, one per pool flight, drawn with a fixed seed for each piece and cell; the bootstrap draws are the same in every cell.

# G Reweighting to the piece

Piece $g$ of G has terms $a_g$ (its MSD sums, or its phase terms) and controls $s$ with terms $b_s$. With $w_g$ the weight of the piece (from the bootstrap) and $w_s$ that of a control,

$$\text{group} = \sum_g w_g\, a_g, \qquad \text{control} = \sum_g w_g\, \frac{\sum_{s\in g} w_s\, b_s}{\sum_{s\in g} w_s}, \qquad \text{total} = \sum_g w_g,$$

over the pieces with positive weight and at least one control of positive weight. Each piece counts once on both sides, whatever the number of its controls: the control side has the composition of the pieces. The MSD at each lag is the mean group/total (resp. control/total), the logarithm and the slopes are taken after the means, and the phase statistics are the ratios $\sum_g w_g \bar r_g / \sum_g w_g \bar q_g$ with $\bar r_g$, $\bar q_g$ the means of $r$ and $q$ over the controls of the piece.

*Example.* Two pieces with terms $a = 5$ and 7, the first with the controls 2 and 4, the second with the control 6, all weights 1. The group side is 6.0; the control side is the mean over the pieces of the means of their controls, 4.5, against 4.0 for the plain mean of the three controls, in which the piece with two controls would weigh twice.

# H The bands: a two-way bootstrap of pilots and blocks

*What is random.* The archive holds the flights of some pilots on some days; other seasons would bring other pilots and other days, and every number of this report would change. The flights are not independent of each other. Two flights of one pilot share the pilot: skill, wing, habits. Two flights of one block share the day and the group: a block joins, into connected components, the flights tied by an episode of the grouping or by a site-day (takeoffs within 3 km on the same day, chained). Pilots and blocks are crossed, since a pilot flies in many blocks and a block holds many pilots, so neither contains the other and neither alone describes the dependence.

*The variance.* To first order an estimate is a sum over the units of the data, $\hat\theta - \theta \approx \sum_k \psi_k$: for a mean over $n$ units $\psi_k = (y_k - \bar y)/n$, and a ratio, a slope or a difference of two populations reduce to this form after linearisation. Two terms are correlated when their units share a pilot or a block, so

$$V = \sum_{k,l} \psi_k \psi_l\, \mathbb{1}[p_k = p_l \text{ or } b_k = b_l] = V_P + V_B - V_C,$$

where $V_P$ sums the pairs with the same pilot, $V_B$ those with the same block, and $V_C$ those with the same pilot and the same block (the same cell), which $V_P$ and $V_B$ both count (Cameron, Gelbach and Miller, *J. Bus. Econ. Stat.* 29, 238, 2011). Each term is the variance of a one-way cluster bootstrap. Drawing the pilots with replacement and giving every unit the count of its pilot as weight, the estimate varies over the draws with variance $V_P$; drawing the blocks gives $V_B$, drawing the cells gives $V_C$.

*In practice.* Three bootstraps of 1 000 draws each, with a fixed seed: of the pilots, of the blocks and of the cells. In a draw every unit, of the group or solo, weighs the count of its pilot (block, cell), and every statistic is recomputed as on the data: the weighted means first, then the ratios, the logarithms and the slopes. All the quantities of a section use the same draws, so a difference of two quantities has the band of its own draws. With $V_P$, $V_B$, $V_C$ the variances of an estimate over the three sets of draws,

$$V = \max\big(V_P + V_B - V_C,\ V_P,\ V_B\big), \qquad \text{band} = \hat\theta \pm 1.645\sqrt{V},$$

a 90% band. When the units of one pilot or of one block covary positively, the true $V$ is at least each one-way variance; the maximum keeps the estimate there when noise makes the sum smaller. A band that excludes 0 is a two-sided test at 10%: among many quantities without an effect, one in ten excludes 0 by chance. The bands of the curves (MSD, $H_{\text{loc}}(\tau)$, survival) hold at each lag separately; the survival band is cut to $[0, 1]$.

*Example.* Three flights with terms $\psi = +1$ (pilot 1, block a), $+1$ (pilot 1, block b) and $-2$ (pilot 2, block b). By pilot the sums are 2 and $-2$, so $V_P = 8$; by block 1 and $-1$, $V_B = 2$; by cell 1, 1, $-2$, $V_C = 6$. Then $V = 4$: the sum of $\psi_k\psi_l$ over the pairs that share a pilot or a block, the three squares (6), the pair of pilot 1 twice ($+2$) and the pair of block b twice ($-4$).

*Why not the product of the counts.* Weighting a unit by the count of its pilot times the count of its block (the pigeonhole bootstrap, Owen, *Ann. Appl. Stat.* 1, 386, 2007; the bands of points 2 and 3 in an earlier version of

this report) gives two units the weight covariance $\mathbb{1}[p_k = p_l] + \mathbb{1}[b_k = b_l] + \mathbb{1}[p_k = p_l]\,\mathbb{1}[b_k = b_l]$. Its variance is then $V_P + V_B + V_C$, the pairs of a cell counted three times instead of once; in the example, 16 against 4. A cell is about one flight, since a pilot flies about once a day, so $V_C$ is about the variance of single flights, and the band of the product is too wide.

*The matched comparisons* (Sections 6 and 7). The pieces and their controls are flights of the same data and are drawn together: in a draw a piece enters with the controls drawn with it, each with its count as weight (Appendix G), and it leaves the draw when none of them is drawn, as it would leave a new sample matched by the same rules. Control flights are shared: 914 control flights serve the 1 754 pieces, and $\sum_s K_s^2 / \sum_s K_s = 6.7$, with $K_s = \sum_{g \ni s} 1/m_g$ the weight a control flight $s$ carries over the pieces $g$ it serves. A control flight that is high or low moves every piece it serves at once. Drawing the control flights keeps this; carrying each piece with fixed controls would treat the pieces that share one as independent and give too narrow a band (Abadie and Imbens, *Econometrica* 76, 1537, 2008).

*The check by simulation.* The script `coverage.py` keeps the designs of points 1, 2 and 4 (the same flights, pilots, blocks, pieces, controls and events) and replaces the value of every unit by $y = a_{\text{pilot}} + b_{\text{block}} + c_{\text{flight}} + e_{\text{unit}}$, independent normal effects, with no group effect. It does so under five mixes of the four variances (Table 12), 400 times each, with 400 draws per bootstrap. The standard deviation of the estimate over the replicates is the one the band must reproduce, and a 90% band must hold 0 in 90% of the replicates. The two-way band holds 0 in 88–99% of them, with a standard deviation 0.93 to 1.43 times the true one; its largest value, 1.43, is in point 2 when nothing ties the segments of one control flight, a case where the band is wide, which is the safe side. The bands used before are off: in point 1, blocks of episodes only hold 0 in 59% of the replicates when pilots matter; in point 2, the product of the counts holds it in 99–100%, with a standard deviation 1.5 to 2.7 times the true one.

| | point 1 | | point 2 | | point 4 | |
|---|---|---|---|---|---|---|
| variances | two-way | before | two-way | before | two-way | before |
| unit only (0, 0, 0, 1) | 91 (1.05) | 90 (1.02) | 99 (1.43) | 100 (2.69) | 90 (1.02) | 88 (0.99) |
| flight only (0, 0, 1, 0) | 92 (1.03) | 91 (1.00) | 89 (1.00) | 100 (1.69) | 90 (1.01) | 90 (0.98) |
| half flight, half unit (0.05, 0.15, 0.5, 0.5) | 88 (0.93) | 85 (0.85) | 92 (1.05) | 100 (1.79) | 88 (1.00) | 88 (0.97) |
| pilot strong (1, 0.2, 0.5, 0.3) | 88 (0.94) | 59 (0.53) | 88 (0.94) | 100 (1.51) | 89 (1.00) | 81 (0.78) |
| block strong (0.1, 1, 0.5, 0.3) | 88 (0.93) | 84 (0.86) | 88 (0.98) | 99 (1.57) | 89 (0.99) | 88 (0.97) |

**Table 12:** Share of the replicates (%) whose 90% band holds 0, and in parentheses the mean standard deviation of the band over the true one, for the two-way band of this report and the band used before in each point: blocks of episodes only (point 1), the product of the counts of pilots and blocks (point 2), blocks of episodes and site-days (point 4); all three before used the 5–95% of the draws. Variances of the pilot, block, flight and unit effects in the first column.

*What the band leaves out.* The rules of the matching ($\rho$, cover, calipers), which the robustness lines of Section 6 vary instead; the random windows of the cells without R (Section 7), drawn once; and the dependence on one year, which Appendix I shows.

# I Leaving one year out

For each year of the archive the estimate is repeated with weight 0 on the flights of that year (the same form as a bootstrap weight). The thin bars of Figures 7 and 10 are the minimum and the maximum of these estimates: a result that depends on one year shows as a wide thin bar.

# J The Shapley decomposition

Let $\Delta(C)$, $C \subseteq F = \{R, T, W\}$, be the gap of G against the controls that satisfy only the conditions of $C$. The part of the raw gap that comes from the factor $k$ is

$$\varphi_k = \sum_{C \subseteq F \setminus \{k\}} \frac{|C|!\,(2 - |C|)!}{3!} \left[\Delta(C) - \Delta(C \cup \{k\})\right],$$

with weights $\frac{1}{3}$ for $|C| = 0$, $\frac{1}{6}$ for $|C| = 1$ (two such sets) and $\frac{1}{3}$ for $|C| = 2$: they add to 1. They are the shares of the $3! = 6$ orders of the factors in which $k$ comes after exactly the factors of $C$. Along one order $\Delta(\varnothing) - \Delta(F)$ is a sum of the three differences of consecutive cells, one per factor. Averaging over the six orders, $\sum_k \varphi_k = \Delta(\varnothing) - \Delta(F)$ exactly:

$$\Delta(\varnothing) = \Delta(F) + \varphi_R + \varphi_T + \varphi_W.$$

Every order has the same weight, so the order in which the factors are added does not matter. The identity holds on every bootstrap draw (the maximum gap is 0 to the precision of the run), so the bands of the $\varphi$ and of $\Delta(F)$ come from the same draws.

*Additive example.* If $\Delta(C) = g + \sum_{k \notin C} a_k$, then $\Delta(C) - \Delta(C \cup \{k\}) = a_k$ for every $C$, and $\varphi_k = a_k$. With $g = 0.01$, $a_R = 0.30$, $a_T = 0.05$, $a_W = 0.10$: $\Delta(F) = 0.01$, $\Delta(\varnothing) = 0.46$, and $\varphi_R = 0.30$, $\varphi_T = 0.05$, $\varphi_W = 0.10$. The shares $\varphi_k / \Delta(\varnothing)$ are meaningful only when $\Delta(\varnothing)$ is away from 0.

# K The drift-free estimator

Let a quantity $x$ drift linearly in time by $\delta$ per window, and let the group add $\gamma$ inside the stretch. With four adjacent windows, pre then start at the entry, end then post at the exit,

$$D_{\text{in}} = x_{\text{start}} - x_{\text{pre}} = \gamma + \delta, \qquad D_{\text{out}} = x_{\text{end}} - x_{\text{post}} = \gamma - \delta,$$

because the end window comes before the post window. So $g = (D_{\text{in}} + D_{\text{out}})/2 = \gamma$ and $d_w = (D_{\text{in}} - D_{\text{out}})/2 = \delta$. The assumptions are a drift linear in time and a group effect that is the same on entry and on exit. Each of $D_{\text{in}}$, $D_{\text{out}}$, $g$ and $d_w$ is computed on the events that have the four windows, with the same bootstrap draws.

*Example.* $D_{\text{in}} = 3$ and $D_{\text{out}} = 1$ give $g = 2$ and $d_w = 1$.

# L Figures of the other datasets and cells

| | G | ∅ | R | T | RT | W | RW | TW | RTW |
|---|---|---|---|---|---|---|---|---|---|
| $H_{\text{loc}}(1000\text{ s})$ | 0.906 | 0.908 | 0.896 | 0.909 | 0.899 | 0.901 | 0.898 | 0.899 | 0.899 |
| $H_{\text{loc}}(1840\text{ s})$ | 0.893 | 0.925 | 0.870 | 0.927 | 0.873 | 0.917 | 0.875 | 0.915 | 0.874 |
| time in climb (%) | 29.2 | 38.8 | 26.9 | 39.0 | 26.6 | 37.7 | 26.6 | 37.8 | 26.3 |
| time in search (%) | 15.3 | 9.7 | 13.6 | 9.7 | 14.0 | 10.5 | 14.0 | 10.6 | 14.2 |
| searches ending in a climb (%) | 49.5 | 45.6 | 47.2 | 47.3 | 46.9 | 44.5 | 46.4 | 45.8 | 46.7 |
| climbs per hour | 7.63 | 6.94 | 7.08 | 7.17 | 7.04 | 6.91 | 7.04 | 7.11 | 6.98 |
| climb rate (m/s) | 1.16 | 1.09 | 1.21 | 1.14 | 1.22 | 1.10 | 1.21 | 1.14 | 1.22 |
| glide speed (km/h) | 29.5 | 35.2 | 30.9 | 35.7 | 31.0 | 33.5 | 30.6 | 33.5 | 30.9 |
| height lost per km of glide (m/km) | 69 | 95 | 63 | 94 | 63 | 99 | 63 | 99 | 64 |

**Table 13:** Decomposition, level all, on 1 754 pieces of G (706 for $H_{\text{loc}}(1840\text{ s})$, 866 for the phase statistics): value of G and of the controls of each cell; the cells are named by the conditions they hold (∅ none, R route, T season and hour, W wing). Dataset 1000 for $H_{\text{loc}}(1000\text{ s})$ and the statistics, 2250 for $H_{\text{loc}}(1840\text{ s})$; average first, ratio after, unit the piece.

| | ∅ | R | T | RT | W | RW | TW | RTW |
|---|---|---|---|---|---|---|---|---|
| $H_{\text{loc}}(1000\text{ s})$ | −0.002 | +0.010* | −0.003 | +0.007* | +0.005 | +0.008* | +0.008* | +0.007 |
| $H_{\text{loc}}(1840\text{ s})$ | −0.032* | +0.023* | −0.035* | +0.020* | −0.024* | +0.018* | −0.022* | +0.018* |
| time in climb (%) | −9.6* | +2.3* | −9.8* | +2.6* | −8.6* | +2.6* | −8.6* | +2.9* |
| time in search (%) | +5.6* | +1.7* | +5.6* | +1.3 | +4.8* | +1.3 | +4.7* | +1.1 |
| searches ending in a climb (%) | +3.9* | +2.2 | +2.2* | +2.6 | +5.0* | +3.0* | +3.7* | +2.8 |
| climbs per hour | +0.69* | +0.55* | +0.46* | +0.58* | +0.72* | +0.58* | +0.52* | +0.64* |
| climb rate (m/s) | +0.071* | −0.042* | +0.023 | −0.055* | +0.067* | −0.043* | +0.022 | −0.053 |
| glide speed (km/h) | −5.7* | −1.3* | −6.2* | −1.5* | −4.0* | −1.0* | −4.0* | −1.3* |
| height lost per km of glide (m/km) | −27* | +6* | −26* | +6* | −31* | +6* | −30* | +5* |

**Table 14:** Decomposition, level all: $\Delta(C) = \text{G} - \text{controls of the cell } C$ for the 8 cells (same pieces and units as Table 13); *: the 90% band excludes 0.

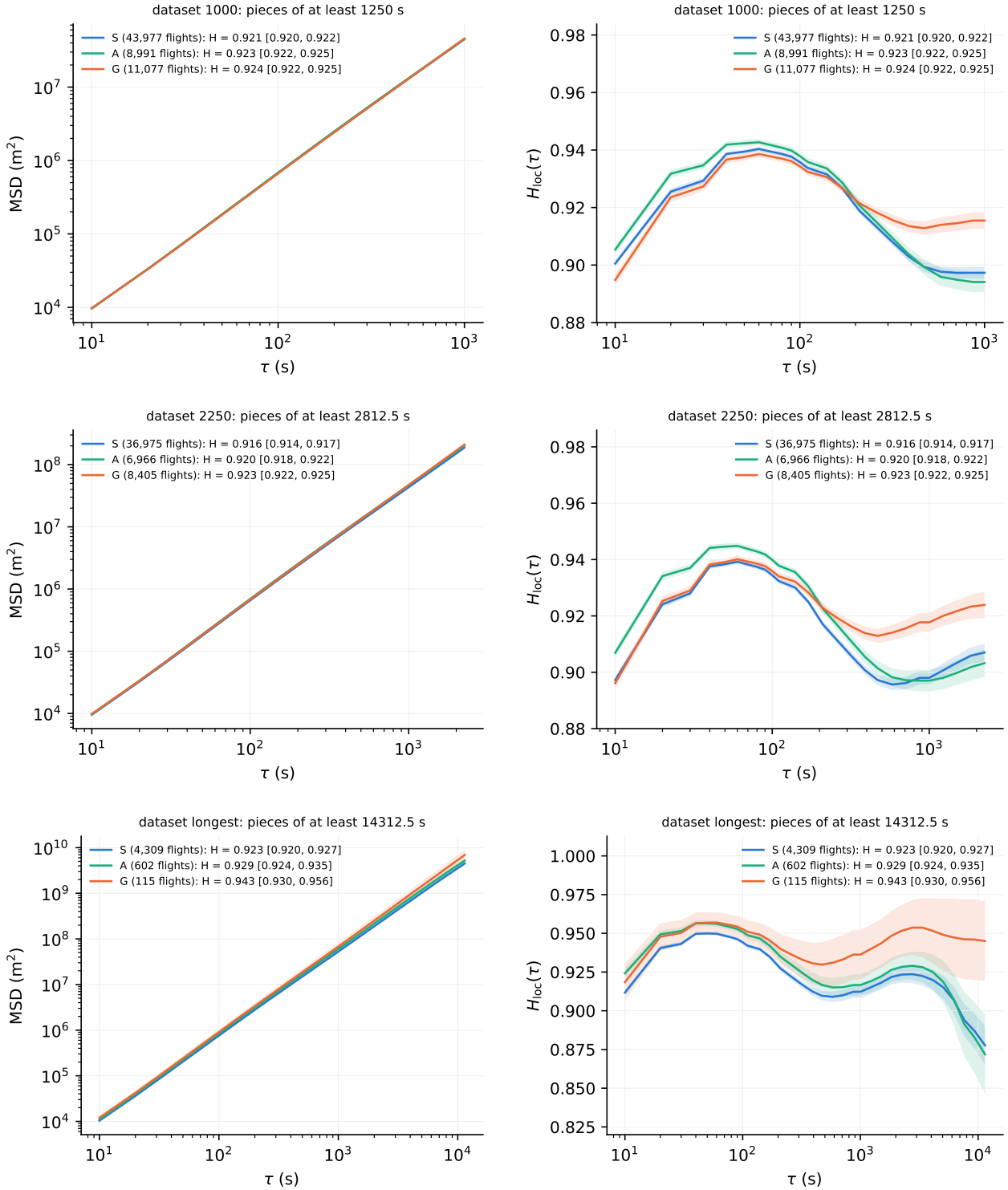



**Figure 14:** Raw, open flights, datasets 1000 (top: S 43 977 flights, A 8 991, G 11 077), 2250 (middle: 36 975, 6 966 and 8 405) and longest (bottom: 4 309, 602 and 115 flights): MSD($\tau$) and $H_{\mathrm{loc}}(\tau)$, ensemble mean over flights, fitted $H$ in the legend, 90% bands.

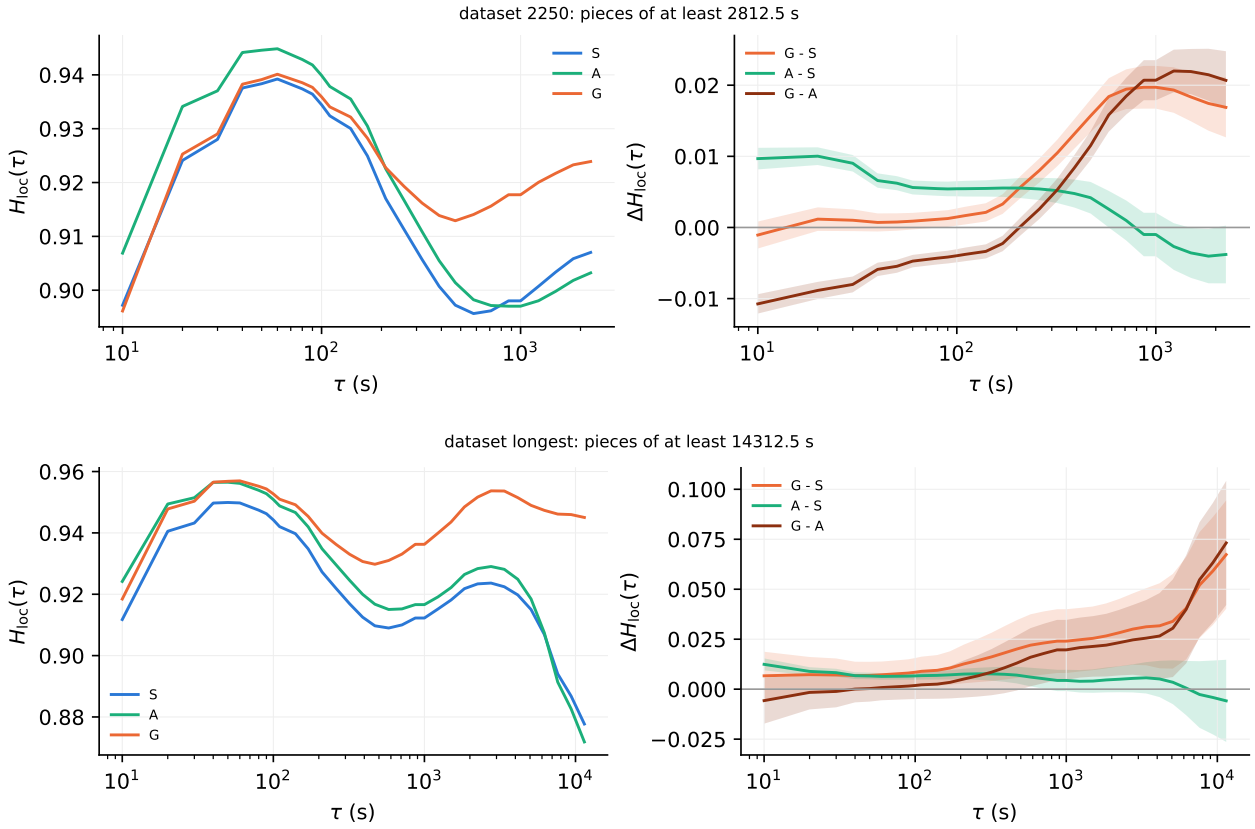



**Figure 15:** Raw, datasets 2250 (top) and longest (bottom): $H_{\mathrm{loc}}(\tau)$ of S, A and G, and the differences at every lag with 90% bands, as Figure 3.

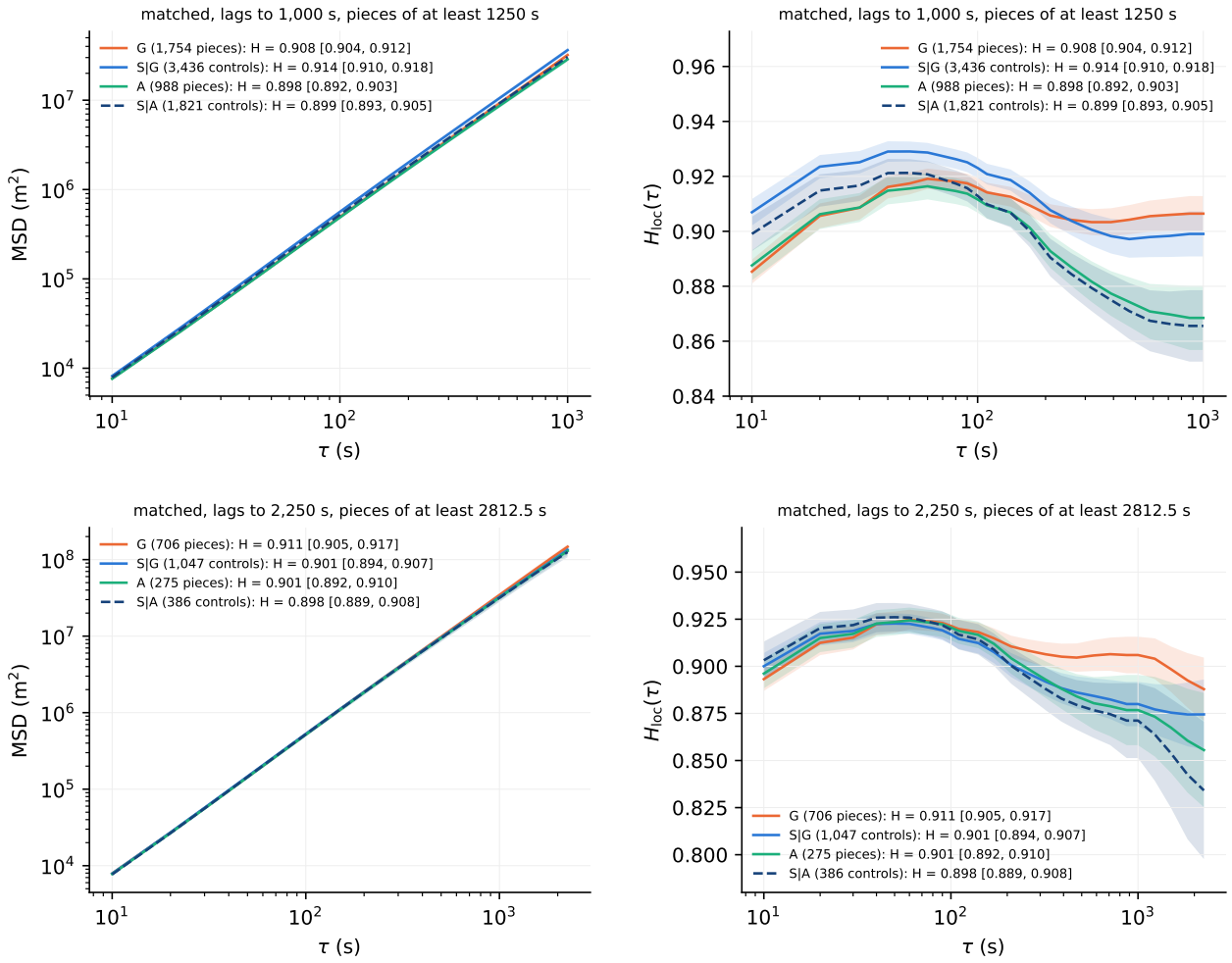



**Figure 16:** Same route, season, hour and wing, datasets 1000 (top) and 2250 (bottom, 706 pieces of G); G, S | G, A, S | A with the pieces or the controls in the legend. MSD($\tau$) and $H_{\mathrm{loc}}(\tau)$, ensemble mean over units reweighted to the piece, fitted $H$ in the legend, 90% bands.

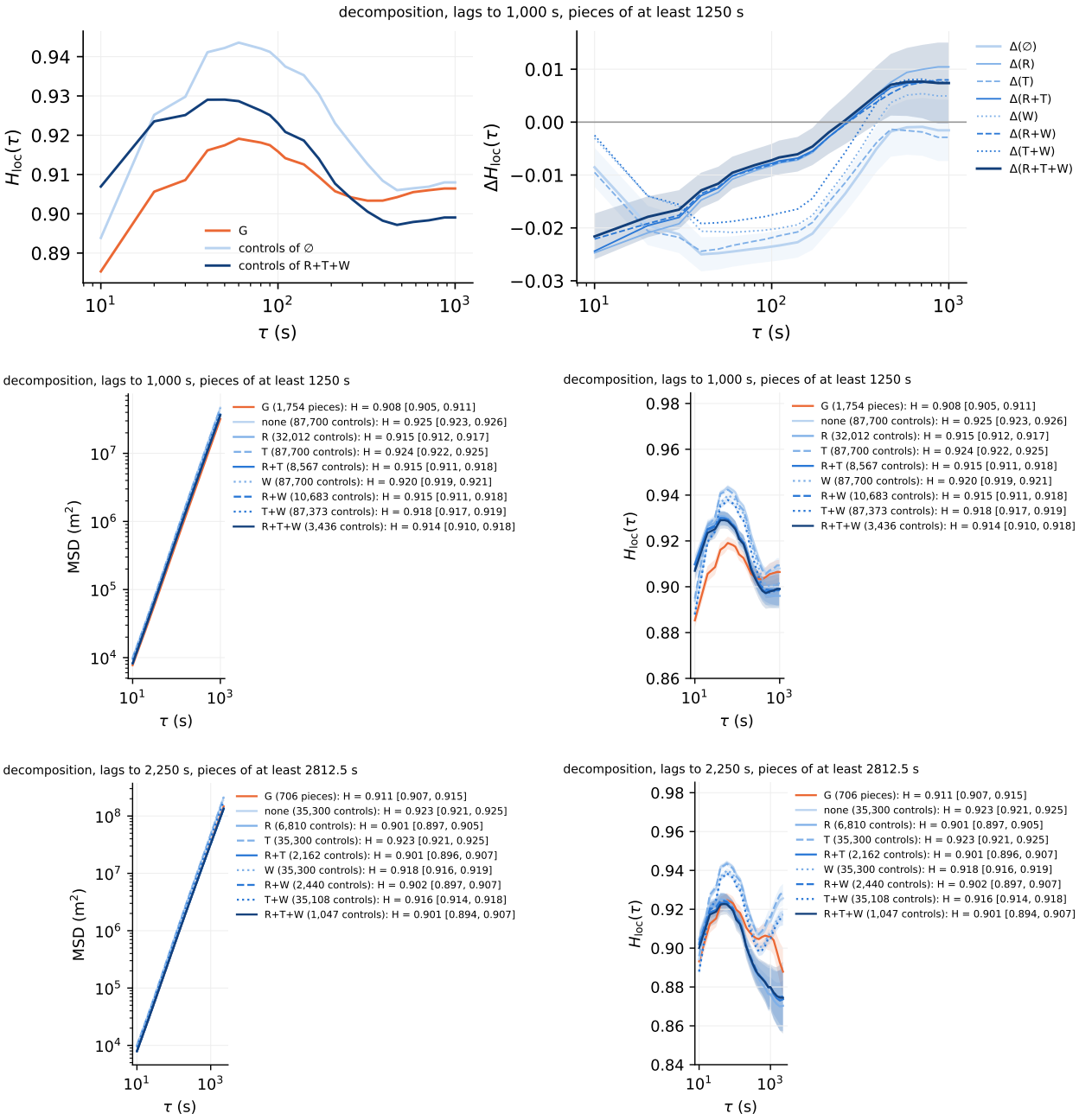



**Figure 17:** Decomposition. Top: dataset 1000 (1 754 pieces), as Figure 11. Middle and bottom: G and the controls of the 8 cells, datasets 1000 and 2250 (706 pieces), the number of controls in the legend, the darker the more conditions; MSD($\tau$) and $H_{\text{loc}}(\tau)$, ensemble mean over pieces, fitted $H$ in the legend, 90% bands.

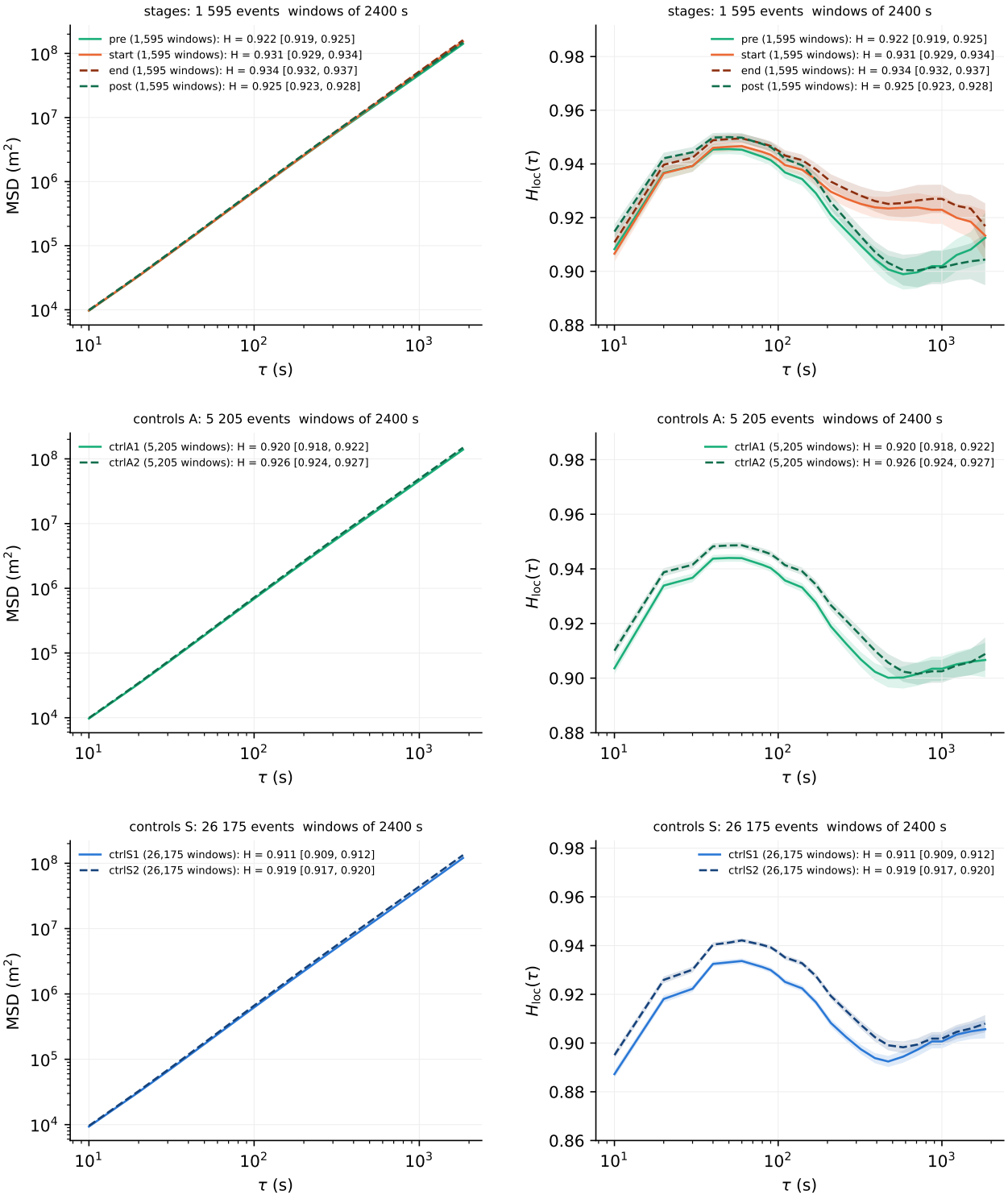



**Figure 18:** Section 8: the four windows of the 1 595 events with the four windows (top), and the control pairs, two consecutive windows of 2 400 s in the solo time of the flights of A (middle, 5 205 events) and in solo flights (bottom, 26 175 events); MSD($\tau$) and $H_{\text{loc}}(\tau)$ over every lag (the window of $H_{\text{loc}}$ is cut only at the last lag), fitted $H$ in the legend, 90% bands.